IVIVC App User Guide

Contents

1 Introduction **2**

2 What the app does **2**

3 Getting started **3**

4 Inputs **3**
- 4.1 Input datasets . . . 4
- 4.2 Analysis approaches and models . . . 4
  - 4.2.1 Approaches . . . 4
  - 4.2.2 Models . . . 5
- 4.3 Preprocessing options . . . 8
  - 4.3.1 Normalization . . . 9
  - 4.3.2 Scaling . . . 9
  - 4.3.3 Interpolation . . . 9
  - 4.3.4 How interpolation is used by each approach . . . 10
- 4.4 Performance metrics . . . 11
- 4.5 Cross-validation options . . . 12
  - 4.5.1 Choosing a cross-validation scheme . . . 13
  - 4.5.2 Cross-validation parameters . . . 14
- 4.6 Grid-search initialization . . . 14
- 4.7 Diagnostic plot display . . . 15

5 Outputs **15**
- 5.1 Relative model evidence . . . 15
- 5.2 Prediction outputs . . . 16
- 5.3 Uncertainty and error bars . . . 17
  - 5.3.1 Approach 1: Direct Correlation . . . 17
  - 5.3.2 Approach 2: Independent Model Fitting . . . 18
  - 5.3.3 Approach 3: Direct Mapping . . . 19

6 Miscellaneous **19**
- 6.1 Software verification and validation . . . 19
- 6.2 Command-line interface . . . 19
- 6.3 Extending the app . . . 19
- 6.4 Data handling . . . 19

Appendices **20**

A How to Interpret Model Performance **20**
- A.1 Model comparability before interpreting metrics . . . 20
- A.2 Metric ranges, direction, and interpretation . . . 20
- A.3 Interpreting goodness of fit and prediction error . . . 21
- A.4 Interpreting relative model evidence . . . 21
- A.5 Interpreting cross-validation performance . . . 22
- A.6 Interpreting residual diagnostics . . . 24

A.7 Interpreting parameter uncertainty . . . 25
A.8 Considering model complexity . . . 26
A.9 Scientific plausibility and intended use . . . 26
A.10 Example model review workflow . . . 26
A.11 Questions to guide model review . . . 26
A.12 Example interpretation statements . . . 28

B Worked Example: Comparing Candidate Models 29
B.1 Purpose of the example . . . 29
B.2 Simulated dataset and analysis settings . . . 29
B.3 Fitted curves . . . 29
B.4 Fit metrics, model complexity, and cross-validation . . . 30
B.5 Residuals versus time and missed structure . . . 31
B.6 Q–Q plots and residual distribution shape . . . 31
B.7 Effect of the number of observations . . . 31
B.8 Fitted-parameter uncertainty . . . 33
B.9 Integrated interpretation . . . 33

C Worked Example: Sampling Times, Noise, and Model Uncertainty 35
C.1 Purpose and setup . . . 35
C.2 What the fitted curves show . . . 35
C.3 Recognizing weak support in the app . . . 37

D Glossary 38

References 41

1 Introduction

Establishing an in vitro-in vivo correlation (IVIVC) can support the evaluation of medical devices that include absorbable materials by relating in vivo changes in material properties over time to corresponding in vitro test results, such as degradation profiles. Potential uses of IVIVCs include evaluating manufacturing or material changes and supporting predictions of in vivo performance when modifying an existing device design. This tool is intended to facilitate IVIVC development. The user supplies paired in vitro and in vivo fitting data; an optional in vitro prediction dataset can also be supplied. The app applies selected preprocessing steps, fits and compares multiple correlation approaches and model families, evaluates selected performance metrics and cross-validation results, generates fitting and prediction plots, and produces report-ready summaries for user review and external documentation.

The app fits empirical relationships between the in vitro and in vivo time courses supplied by the user. The scope includes absorbable polymers that degrade by hydrolysis or enzymatic action, as well as absorbable metals. It does not directly model material chemistry, device geometry, implantation conditions, or degradation mechanisms.

This guide describes the app and offers considerations for reviewing its outputs. The app provides information to support user review of models; it does not select or recommend a model or determine regulatory acceptability. The app and user guide are advisory and do not establish requirements, acceptance criteria, or a prescribed model-selection process. Users are responsible for determining how to document their model-selection rationale and other analysis decisions, consistent with the intended use and applicable credibility considerations.

For general background on credibility assessment and reporting of computational modeling studies in medical device submissions, see FDA guidance on computational modeling credibility and reporting [1, 2].

For additional discussion of model performance, cross-validation results, uncertainty, residual diagnostics, and model selection considerations, see Appendix A. For an example of interpreting model performance, see Appendix B. For definitions of commonly used terms, see Appendix D.

2 What the app does

- Applies selected normalization and scaling, and uses interpolation where required for paired-data approaches

and mappings.

- Fits user-selected IVIVC approaches and model families to paired in vitro and in vivo data.
- Calculates selected goodness-of-fit, model-comparison, and cross-validation metrics.
- Generates fitting, comparison, and prediction plots with uncertainty displays where supported.
- Creates summaries for user-selected approaches or models for downstream review and documentation.

3 Getting started

The following provides a brief orientation for a first run of the app. The settings described here as defaults are software defaults only; they do not establish preferred analysis choices or regulatory expectations. Detailed descriptions and interpretation considerations are provided in the sections that follow.

Minimum input. A fitting dataset is required with four columns containing in vitro time, in vitro response, in vivo time, and in vivo response. A separate two-column in vitro prediction dataset is optional and is needed only when generating predictions from fitted or mapped relationships.

From setup to results. On the setup page, upload the fitting dataset and, if desired, a prediction dataset. Select at least one analysis approach and, for model-based approaches, at least one candidate model. Review or modify the preprocessing, model-fitting, performance-metric, cross-validation, and diagnostic-plot options as applicable, then select **Analyze**. The app processes the uploaded data, performs the selected analyses, and opens the results page, where results are organized by processing, fitting, model comparison, and prediction outputs as applicable.

Current software defaults. No analysis approaches or candidate models are selected initially. No normalization or scaling is selected. Shared-grid interpolation is selected for approaches that use matched quantities. The default displayed metrics are R^2, RMSE, and AIC_c. Shuffle-split cross-validation is selected by default. The available CV-fit-variability, raw-residual, and Q–Q diagnostic plots are not enabled by default. These defaults provide a starting configuration and can be changed to match the dataset and analysis purpose.

Main outputs. After analysis, the results include the processed data used by each approach, fitted or mapped relationships, selected performance and cross-validation summaries, diagnostic plots, model-comparison information where applicable, and prediction outputs when a prediction dataset is supplied. User-selected results can also be assembled into a downloadable summary report.

4 Inputs

The IVIVC app requires paired in vitro and in vivo degradation data, along with user-selected analysis settings that define how the data are processed, modeled, and summarized. At a high level, the inputs include:

- a spreadsheet containing in vitro and in vivo time-course degradation data for fitting correlations;
- optionally, a spreadsheet containing in vitro data from which to generate predictions using fitted correlations;
- selection of one or more IVIVC modeling approaches;
- selection of candidate models to evaluate, where applicable;
- optionally, preprocessing options, such as normalization, interpolation, and scaling;
- selected performance metrics to display in the results and reports;
- optionally, cross-validation settings for applicable approaches;
- grid-search settings for model initialization, for which the app provides default values; and
- optionally, advanced analysis plotting settings;

The sections below provide additional detail on input file structure, required data columns, preprocessing options, model selection, and optional analysis settings.

4.1 Input datasets

The app uses separate fitting and prediction datasets. The fitting dataset is required because it provides the paired in vitro and in vivo data used to establish the IVIVC relationship. The prediction dataset is optional and is used only when applying a fitted relationship to new in vitro observations in order to predict the corresponding in vivo behavior.

In this guide, **response value** means the measured y-axis value that changes with time, such as mass remaining, molecular weight, a mechanical property (e.g., tensile or burst strength), or another quantitative degradation-related endpoint. In the input files, these are the in vitro and in vivo value columns.

When interpreting an IVIVC, consider measurement reliability, sample size, comparability of the in vitro and in vivo measurements, and experimental limitations such as incomplete material recovery or contamination, because a good statistical fit does not rule out systematic measurement error.

Input	Description
Fitting dataset	Required four-column dataset used to establish the in vitro/in vivo relationship. Columns are interpreted as in vitro time, in vitro response value, in vivo time, and in vivo response value.
Prediction dataset	Optional two-column in vitro dataset used after fitting to generate in vivo predictions. Columns are interpreted as in vitro time and in vitro response value.

Supported formats are CSV and Excel (.xls, .xlsx). For Excel workbooks, select the sheet containing the relevant data. For CSV files, the first columns are read directly in the order described above.

4.2 Analysis approaches and models

In this app, an analysis **approach** is the overarching method used to establish or evaluate an in vitro/in vivo correlation. A **model** is the mathematical form used within an approach when an equation is needed to describe the relationship.

4.2.1 Approaches

Approaches differ in how the in vitro and in vivo time courses are related to one another. Approach numbers identify the methods and do not indicate a ranking or preference.

Approach	Description	May be relevant when	Available models
Approach 1: Direct Correlation	Approach 1 directly relates the in vitro and in vivo times associated with the same response values after preprocessing (i.e., in vivo time is modeled as a function of in vitro time). When the datasets do not contain the same response values, the app can interpolate time at shared response values to create the paired times used for fitting.	This approach may be relevant when a key analytical question concerns how in vitro time at a given response corresponds to in vivo time at that response.	Double Exponential, Exponential, Linear, Linear (zero intercept), Power

Approach	Description	May be relevant when	Available models
Approach 2: Independent Model Fitting	Approach 2 fits the same model family (e.g., exponential decay) separately to the preprocessed observed in vitro and in vivo datasets. The two datasets do not need matching time points or response values, so interpolation is not used for Approach 2 fitting.	This approach may be relevant when the two datasets were measured on different time schedules, when each dataset is expected to follow the same type of degradation curve, or when a specific model form is scientifically meaningful for both datasets.	Double Exponential, Exponential, Linear + Power Decay, Linear + Power Decay (Normalized), Mass Loss (Gompertz), Mass Loss (Gompertz, Normalized), Mass Loss (One-Phase), Mass Loss (One-Phase, Normalized), Mass Loss (Surface Erosion), Mass Loss (Surface Erosion, Normalized), Mass Loss (Two-Phase), Mass Loss (Two-Phase, Normalized), Stretched Exponential
Approach 3: Direct Mapping	Approach 3 uses direct interpolation-based mappings between datasets rather than fitting a parametric model. Response values interpolated at shared times are used for response-value ratios over time, while times interpolated at shared response values are used for time ratios across response-value ranges.	This approach may be relevant for exploratory comparisons, for datasets where a simple equation does not describe the observed pattern well, or when the user wants a direct data-based mapping rather than a fitted model.	No parametric model selection

When a prediction dataset is supplied, each approach uses the prediction data differently. See Section 5.2 for the available prediction methods and outputs.

For Approach 2 models where a finite value can be calculated, the app summarizes each fitted decay curve with an **integral time constant**. For a fitted model $f(t)$, the time constant is defined as

$$\tau = \frac{1}{f(0)} \int_0^\infty f(t)\, dt,$$

which is the area under the fitted curve after normalizing its fitted value at time zero to 1 [3]. It provides a single characteristic time scale for comparing fitted in vitro and in vivo profiles; a larger value generally corresponds to slower overall decay. For a simple exponential decay, the integral equation yields the time constant. The fitted time constants are also used by one of the optional Approach 2 prediction methods described in Section 5.2.

4.2.2 Models

Candidate models are the mathematical equations available for fitting within the applicable approaches. Different model forms represent different assumptions about the shape of the degradation relationship, such as linear behavior, power-law behavior, or sigmoidal transition behavior.

Model	Used in	Functional form	Notes
Double Exponential	Approach 1, Approach 2	$y = ae^{bx} + ce^{dx}$	Sum of two exponential terms. It can represent profiles with two apparent rates, but the added flexibility may require more data to estimate stably.

Model	Used in	Functional form	Notes
Exponential	Approach 1, Approach 2	$y = ae^{bx}$	Two-parameter exponential relationship. It can describe monotonic change that accelerates or decelerates exponentially over the fitted range.
Linear	Approach 1	$y = a + bx$	Straight-line empirical relationship with an intercept and slope. It is appropriate when the in vitro/in vivo relationship is approximately linear over the fitted range.
Linear (zero intercept)	Approach 1	$y = ax$	Straight-line empirical relationship constrained to pass through zero. It may be useful when a zero in vitro value should correspond to a zero in vivo value under the selected preprocessing and response scale.
Linear + Power Decay	Approach 2	$y = y_0 \max\{1 - kx - (x/t_f)^n, 0\}$	Empirical remaining-response model combining linear and power-law loss, with the initial response fitted and negative remaining values set to zero. For n > 1, the power term can describe accelerating loss. Requires nonnegative time; k >= 0, tf > 0, n > 0, and y0 >= 0.
Linear + Power Decay (Normalized)	Approach 2	$y = \max\{1 - kx - (x/t_f)^n, 0\}$	Linear-plus-power remaining-response model for data already on a 0-1 scale. The initial response is fixed at 1, leaving the linear rate, power-term time scale, and exponent to be fitted. Requires nonnegative time; k >= 0, tf > 0, and n > 0.
Mass Loss (Gompertz)	Approach 2	$y = y_0 - A\exp\{-\exp[(er_{max}/A)(t_{lag} - x) + 1]\}$	Mass-remaining model based on the modified Gompertz curve. It is intended for asymmetric sigmoidal mass-loss profiles with an apparent lag or plateau, a maximum loss-rate region, and a terminal loss extent.

Model	Used in	Functional form	Notes
Mass Loss (Gompertz, Normalized)	Approach 2	$y = 1 - \exp\{-\exp[er_{max}(t_{lag} - x) + 1]\}$	Less-flexible modified Gompertz mass-remaining model for responses already normalized from 0 to 1. The baseline response and ultimate loss magnitude are fixed at 1, leaving only the maximum loss-rate and lag-time parameters to be fitted.
Mass Loss (One-Phase)	Approach 2	$y = y_0 - A/(1 + e^{-(x-t_{50})/s})$	Mass-remaining model with a delayed sigmoidal loss phase. It is intended for mass-loss profiles that begin constant and show a later drop. This is a scaled, shifted, and inverted logistic (sigmoid) function.
Mass Loss (One-Phase, Normalized)	Approach 2	$y = 1 - 1/(1 + e^{-(x-t_{50})/s})$	Less-flexible one-phase mass-remaining model for responses already normalized from 0 to 1. The baseline response and delayed-loss magnitude are fixed at 1, leaving only the transition midpoint and width to be fitted. This is a shifted and inverted logistic (sigmoid) function.
Mass Loss (Surface Erosion)	Approach 2	$y = y_0 \max(1 - kx, 0)^n$	Mass-remaining model for surface-controlled erosion. The fitted exponent n allows the profile shape to adapt to slab-like, cylindrical, spherical, or empirical geometry-like erosion behavior.
Mass Loss (Surface Erosion, Normalized)	Approach 2	$y = \max(1 - kx, 0)^n$	Less-flexible surface-erosion mass-remaining model for responses already normalized from 0 to 1. The response at time zero is fixed at 1, leaving the erosion-rate and geometry/shape parameters to be fitted.
Mass Loss (Two-Phase)	Approach 2	$y = y_0 - B(1 - e^{-k_b x}) - D/(1 + e^{-(x-t_{50})/s})$	Mass-remaining model with an optional early burst-loss phase followed by a delayed sigmoidal loss phase. It is intended for mass-loss profiles that may show an initial drop, an intermediate plateau, and a later drop.

Model	Used in	Functional form	Notes
Mass Loss (Two-Phase, Normalized)	Approach 2	$y = 1 - B(1 - e^{-k_b x}) - D/(1 + e^{-(x - t_{50})/s})$	Less-flexible two-phase mass-remaining model for responses already normalized from 0 to 1. The baseline response is fixed at 1, while the early and delayed loss magnitudes (B and D) and their timing/rate parameters remain fitted.
Power	Approach 1	$y = ax^b$	Power-law empirical relationship. It can describe nonlinear monotonic relationships where the response changes as a fitted power of time or the mapped predictor.
Stretched Exponential	Approach 2	$y = ae^{bx^c}$	Exponential relationship with a fitted stretching exponent. It can describe non-single-rate behavior, but the additional shape parameter can increase sensitivity when data are sparse.

Mass loss model notes

The surface erosion mass loss form is based on geometry-controlled surface erosion concepts from the erodible slab, cylinder, and sphere model [4]. The one-phase mass loss model uses a logistic/sigmoidal transition to represent a delayed loss phase [5]. The two-phase mass loss model combines an early burst-like loss term with a delayed sigmoidal loss phase. Burst and delayed/multiphase behavior in degradable polymer systems is discussed in the polymer erosion and PLGA release literature [6, 7, 8]. The Gompertz mass loss model is a mathematical adaptation of the modified Gompertz form originally developed for microbial growth kinetics [9]. Modified Gompertz functions have also been used as empirical models of polymer biodegradation weight-loss profiles [10]. Its inclusion in the app provides an empirical sigmoidal option with lag and maximum-rate parameters rather than a mechanistic model specific to absorbable medical-device polymers.

The mass loss and linear-plus-power models also have less-flexible normalized versions for response data already expressed on a 0–1 scale. These versions fix the baseline response level at 1 rather than estimating it. For the one-phase and Gompertz forms, the total loss magnitude is also fixed at 1, which fixes the upper and lower asymptotes at 1 and 0. The normalized surface-erosion and linear-plus-power forms fix the response at time zero at 1. The normalized two-phase form fixes the baseline level at 1 but retains separate fitted early and delayed loss magnitudes (B and D). These constraints reduce the number of fitted parameters and assume that the analyzed response already matches the fixed 0–1 response scale. Selecting a normalized model does not itself normalize the data; a preprocessing normalization or input data already on the 0–1 scale is therefore needed for those fixed-response assumptions to hold.

The Linear + Power Decay form is an empirical option for remaining mass, strength, or another declining response. Its t_f parameter scales the power term; when $k > 0$, the curve reaches zero before t_f. When $n = 1$, the two loss terms combine into one slope, so k and t_f cannot be estimated separately. Time must be nonnegative after preprocessing. The flat zero tail does not identify a unique time at zero response, and response-value-ratio predictions are undefined where the fitted in vitro response is zero. Uncertainty near the zero cutoff needs particular care because clipping introduces a corner in the fitted curve.

4.3 Preprocessing options

Preprocessing prepares the uploaded data for comparison, fitting, and prediction. Normalization expresses response values relative to a reference value so datasets can be compared on a common relative scale. Scaling changes the

numerical scale of time or response values without changing the underlying pattern. Interpolation is used when an approach requires paired in vitro and in vivo quantities that were not measured at exactly matching positions. The app constructs both response-at-time and time-at-response representations; these are used differently by Approaches 1 and 3 and are not used for Approach 2 fitting.

4.3.1 Normalization

Normalization changes how response values are expressed without adding or removing observations. Initial-value normalization expresses each profile relative to its first response value, whereas min-max normalization expresses each profile relative to its observed range; either transformation changes the response scale used for subsequent fitting, mapping, and metrics.

Option	Internal key	Description
Min-max normalization	min_max	Converts response values to a 0-1 scale using the observed minimum and maximum within each dataset.
Initial-value normalization	initial_value	Expresses each response value relative to the first response value in the corresponding dataset.

4.3.2 Scaling

Scaling changes the numerical representation of time or response values while preserving the observations themselves. The available base-10 logarithmic transformations compress wide numerical ranges and are applied only when the relevant processed values are positive; otherwise the app retains the unscaled values and reports the fallback.

Option	Internal key	Description
Log10 time scaling	log_x	Uses a base-10 log scale for time values before fitting or interpolation. All time values must be greater than zero. If this scaling is selected but the analyzed data contain a zero or negative time, the app leaves time unscaled, continues the analysis, and reports the fallback in Data Processing.
Log10 response scaling	log_y	Uses a base-10 log scale for response values before fitting or interpolation. All response values must be greater than zero after normalization. If this scaling is selected but the processed responses contain a zero or negative value, the app leaves response unscaled, continues the analysis, and reports the fallback in Data Processing.

4.3.3 Interpolation

Interpolation estimates quantities between measured observations when a paired-data approach requires matching time or response positions. The available options differ in how the shared grids are constructed; interpolated points are estimates derived from the observed data rather than additional experimental observations.

Option	Internal key	Description
No interpolation	none	Uses only exact matches from the observed data. No response-at-time or time-at-response estimates are added, so unmatched positions remain unavailable for Approaches 1 and 3. Approach 2 uses observed points regardless of this setting.

Option	Internal key	Description
Shared-grid interpolation	default	Uses linear interpolation in both directions: response values are estimated at the shared set of observed time points, and times are estimated at the shared set of observed response values. Estimates outside the measured overlap are unavailable. Approach 2 does not use these interpolated points for fitting.
Densified interpolation	alternative	Adds intermediate grid points in both time and response-value directions before linear interpolation, producing denser paired mappings for Approaches 1 and 3. The added points are estimates rather than additional experimental observations. Approach 2 does not use them for fitting.

The app preserves observed time zero as $t = 0$ in processed data, fitting, plots, and downloads rather than replacing it with a small positive value. Because $\log_{10}(0)$ is undefined, Log10 time scaling cannot be applied when any analyzed time value is zero or negative. In that case, the app continues the analysis with unscaled time and displays a scaling-fallback warning in Data Processing. Similarly, if Log10 response scaling is selected but the processed response values contain zero or negative values, the app continues with unscaled response values and reports the fallback. The preprocessing table reports the scaling that was actually applied. Model-specific numerical/domain limitations at zero are handled during fitting rather than by changing the supplied time values. For Approach 3, an in vivo/in vitro time ratio with an in vitro time of zero is undefined and is reported as unavailable, although a prediction observed at time zero can still map to time zero.

4.3.4 How interpolation is used by each approach

When interpolation is selected, the app constructs two complementary representations of the preprocessed fitting data:

- **Interpolation by time (response at shared times):** for a shared set of time points, the app estimates the corresponding in vitro and in vivo response values. This representation is used by Approach 3 for response-value mappings over time.
- **Interpolation by response value (time at shared values):** for a shared set of response values, the app estimates the corresponding in vitro and in vivo times. This representation is used by Approach 1 and by the time-mapping part of Approach 3.

The Data Processing results tab shows these two representations separately so that the points used for each mapping can be reviewed directly. With the shared-grid interpolation option, the shared grids are based on the union of observed time points and the union of observed response values, respectively. Densified interpolation adds intermediate grid points in both directions before interpolation. Estimates outside the overlapping data range are not used.

Approach 1: Direct Correlation. Approach 1 relates the in vitro and in vivo *times associated with the same response value*. If the two datasets contain exactly matching response values, those observed pairs can be used directly. Otherwise, interpolation by response value estimates the time at common response values so that the in vitro and in vivo times can be paired for fitting. Matching the original experimental time schedules is not required because the fitted relationship is in vivo time as a function of in vitro time at equivalent response states.

Approach 2: Independent Model Fitting. Approach 2 fits the selected model family independently to each preprocessed observed dataset. The in vitro and in vivo time points do not need to match, and the response values do not need to match, so no alignment by interpolation is required. The app therefore excludes interpolated points from Approach 2 fitting. Interpolated values are deterministic estimates from the measured observations rather than additional independent observations; treating them as fitting data could change the apparent sample size, fit metrics, cross-validation inputs, residual degrees of freedom, and covariance-based parameter uncertainty without adding new experimental information. Normalization and scaling, when selected, are still applied before the independent fits.

Approach 3: Direct Mapping. Approach 3 does not fit a parametric model and instead uses both interpolation directions. Response values at shared times are used to calculate response-value ratios or mappings over time.

Times at shared response values are used to calculate time ratios or mappings across the response range. Both representations are therefore relevant when Approach 3 is selected.

With **No interpolation**, Approaches 1 and 3 can use only the matching points that are actually present in both datasets. If the datasets have few or no matching times or response values, there may be too few points for the comparison. Interpolation can provide additional matched points.

Interpolation by response value assumes that each response value corresponds to a single time. If the response rises and falls over the analyzed range, the same response value may occur at more than one time, making this mapping unclear. Review the by-response-value interpolation plot to confirm the results appear consistent with the data before using this mapping.

4.4 Performance metrics

Performance metrics are summary measures that help describe how closely a fitted model matches the observed data (e.g., R^2), how large the prediction errors are (e.g., RMSE), and, for some metrics, how well a model balances fit quality with model complexity (e.g., information criteria). The following performance metrics are included in the app.

Metric	Internal key	Preferred direction	Description
R^2	r_squared	Higher is better	Coefficient of determination. Higher values indicate that the fitted curve follows more of the observed response pattern.
Adjusted R^2	adjusted_r_squared	Higher is better	R^2 adjusted for the number of fitted parameters. It can decrease when extra parameters do not improve the fit enough to justify the added complexity.
MSE	mse	Lower is better	Average squared difference between observed and fitted values. Lower values indicate smaller model errors, but the squared units can be harder to interpret directly.
RMSE	rmse	Lower is better	Typical model error in the same response units as the fitted data. Lower values indicate predictions closer to the observations.
MAE	mae	Lower is better	Average absolute model error in the same response units as the fitted data. Lower values indicate predictions closer to the observations.
NRMSE (mean)	mnrmse	Lower is better	RMSE divided by the response mean. Lower values indicate smaller error relative to the typical response magnitude.
NRMSE (std. dev.)	nrmse	Lower is better	RMSE divided by the response standard deviation. Lower values indicate smaller error relative to the amount of variation in the response.
AIC	aic	Lower is better	Akaike Information Criterion. AIC compares fitted models by considering both fit quality and the number of fitted parameters. Among comparable fitted models, a lower AIC value generally indicates better empirical support for a model that fits the data without unnecessary complexity. Absolute AIC values, including negative values, are not standalone quality grades.

Metric	Internal key	Preferred direction	Description
AIC_c	aicc	Lower is better	Corrected Akaike Information Criterion. AIC_c is AIC with an added small-sample correction that becomes increasingly relevant when the number of fitted observations is limited relative to the number of fitted parameters. See the user guide for additional information. Among comparable fitted models, a lower AIC_c value generally indicates better empirical support for a model that balances fit quality and complexity; AIC_c is undefined when there are too few observations relative to the number of parameters.
BIC	bic	Lower is better	Bayesian Information Criterion. BIC compares fitted models by considering both fit quality and the number of fitted parameters, usually applying a stronger penalty for extra parameters than AIC. Among comparable fitted models, a lower BIC value generally indicates better empirical support for a model that fits the data without unnecessary complexity. Absolute BIC values, including negative values, are not standalone quality grades.

For additional background on common error metrics, scale-dependent accuracy measures, and adjusted R^2, see references [11, 12, 13].

AIC_c adds a small-sample correction to AIC that is relevant when the ratio of fitted observations to estimated parameters is small. A commonly cited rule of thumb is $n/k < 40$, where n is the number of fitted observations and k is the number of model parameters [14]; this condition is common in typical IVIVC datasets.

By default, the app selects R^2, RMSE, and AIC_c. This default set reflects the tool developers' experience with representative IVIVC datasets and is not intended as a universal or required metric set. Additional metrics are available when they provide useful context for a specific analysis. The default set gives three complementary views of model performance. R^2 is widely known and describes how well the fitted curve follows the observed response pattern. RMSE reports typical error in the original response units, making it easier to judge whether the error is practically large or small. AIC_c supports comparison among fitted candidate models by balancing fit quality with a penalty for unnecessary model complexity, with a correction that is useful for smaller datasets.

4.5 Cross-validation options

Cross-validation provides an estimate of the model's ability to predict observations not used for fitting. The app does this by repeatedly setting aside part of the dataset, fitting the model to the remaining observations, and comparing model predictions with the observations that were set aside. When the cross-validation error is similar to the error from fitting the full dataset, this suggests the fitted relationship is reasonably stable.

In this app, CV is intended as a **prediction-error diagnostic** and sensitivity check, not as a confidence interval or parameter uncertainty estimate. If the error from the model fitted to the full dataset is similar to the CV error, that provides reassurance that the fitted relationship is not overly dependent on a particular subset of observations. If the CV error is much worse, the fitted relationship might be sensitive to the available data, the selected splitting scheme, or the number and placement of time points.

An optional **CV fit variability (mean ± SD)** plot provides a complementary view of that sensitivity. For every CV split that fits successfully, the app evaluates the model fitted to the training subset over the full displayed x-range. At each x-value, it calculates the mean fitted response and standard deviation across the CV fits. The plot shows the

full-data model fitted to all observations, the mean CV-training fit, and a shaded mean ± one-standard-deviation band. A wider band indicates that the fitted curve changes more depending on which observations are included in the training data. This band summarizes variation among CV fits; it is not a confidence interval, prediction interval, or parameter-uncertainty interval. Its appearance depends on the selected CV splitting scheme and the number of successful CV fits.

For time-course data, the splitting scheme matters because nearby observations can contain related information, and early, middle, and late time points can represent different parts of the degradation process. Setting aside a random subset is a general check of whether the fitted curve captures the overall response pattern, but it is less time-aware. Setting aside one time point mainly asks whether the model can recover an individual omitted observation. Setting aside a contiguous block asks whether the model can bridge a missing region of the profile.

Shuffle split is the default CV scheme because it provides a general robustness check of whether the selected curve form captures the overall degradation or correlation pattern. It is not a time-aware forecasting test. Leave-contiguous-block-out may be informative when sensitivity to a missing time region is a primary concern; leave-one-timepoint-out may be informative when the dataset is too small to set aside larger groups of observations. When results change substantially across CV schemes, the fitted plots, residual diagnostics, parameter uncertainty, model complexity, and scientific plausibility provide additional context rather than treating CV as a pass/fail result.

The default CV settings reflect the tool developers' experience with representative IVIVC datasets and are not intended to be universally applicable. Alternative settings may be appropriate depending on the characteristics of the specific dataset and analysis goals.

For background on cross-validation procedures and their use in model assessment and selection, including issues that arise with time series, see references [15, 16, 17, 18].

4.5.1 Choosing a cross-validation scheme

The table below summarizes the schemes available in the app. The "CV error describes" column explains what kind of prediction sensitivity each scheme is most directly measuring.

Scheme	CV error describes	Useful when	Main caution
Shuffle split	Prediction error for repeated random subsets of observations set aside from fitting.	Default general robustness check for whether the fitted curve captures the overall response pattern.	Not time-aware; can appear overly reassuring when nearby time points carry similar information.
Leave-one-timepoint-out	Prediction error when one time point is omitted at a time; sensitivity to setting aside one observation.	Useful for very small datasets because each repeated fit still uses nearly all of the available data.	Can be a gentle test when neighboring time points have similar response values to the omitted point.
Leave-contiguous-block-out	Prediction error when a block of adjacent time points is omitted; sensitivity to a gap in the observed profile.	Primary time-aware sensitivity check, especially when a missing time region is a practical concern.	Larger blocks make the test harder but can leave too few points for the model to fit stably. Smaller blocks retain more observations for each repeated fit.

Scheme	CV error describes	Useful when	Main caution
None	No cross-validation is performed; only the model fitted to the full dataset is summarized.	Useful when there are too few observations to split meaningfully, repeated fits are numerically unstable, or the goal is simply to examine the fitted model itself.	Removes one diagnostic tool. When CV is skipped, other outputs like fitted plots, residuals, full-fit metrics, parameter diagnostics, and scientific plausibility provide the remaining model-review context.

4.5.2 Cross-validation parameters

Parameter	Internal key	Description
Cross-validation scheme	cv_scheme	Selects the splitting method used to estimate cross-validation performance for fitted parametric models. Shuffle split is the default general robustness check.
CV splits/block size	cv_n_splits	For Shuffle split, the number of repeated random train/test splits. For Leave-contiguous-block-out, the number of adjacent time points set aside per split. The setup page does not enforce a maximum block size because the uploaded data are not read until Analyze is selected; larger blocks may leave too few observations for stable fitting. This setting is not used for Leave-one-timepoint-out or None.
Shuffle split test size	cv_test_size	The fraction of observations used as test data in each Shuffle split repeat. Smaller values leave more observations for fitting; larger values make the test more demanding but leave fewer observations for training.
CV random seed	cv_random_state	Seed used to make randomized cross-validation splits reproducible. This applies to Shuffle split.

4.6 Grid-search initialization

When the app fits a mathematical curve, it needs starting values for the model parameters. Poor starting values can lead to an unstable or unreasonable fit, especially for nonlinear models. Grid-search initialization reduces this risk by trying many candidate starting-value sets before nonlinear least-squares fitting.

The app evaluates those candidate starts, chooses the most promising one, and then runs the final curve-fitting step from that starting point. This improves robustness, but it does not establish that the fit is the global optimum.

The app provides default grid-search settings as its initial configuration. Options for addressing unstable fitting include increasing the number of random starts, adjusting the candidate starting-value range, changing the random seed, or evaluating whether the selected model form describes the data adequately.

Parameter	Internal key	Description
Grid-search random starts	grid_search_num_points	Number of candidate starting parameter sets evaluated before nonlinear least-squares fitting. Higher values may improve robustness but increase runtime.

Parameter	Internal key	Description
Grid-search CPU cores	grid_search_num_cores	Number of CPU cores used for the starting-point search. Only available via command-line interface.
Grid-search parameter minimum	grid_search_param_min	Lower bound for randomly sampled candidate starting parameter values.
Grid-search parameter maximum	grid_search_param_max	Upper bound for randomly sampled candidate starting parameter values.
Grid-search random seed	grid_search_random_state	Seed used to make randomized starting-point sampling reproducible. Changing the seed changes the sampled candidates while keeping the same bounds and number of starts.

4.7 Diagnostic plot display

The CV fit variability plot, raw residual plots, and normal quantile-quantile (Q-Q) plots are not selected by default in the Advanced Analysis section to produce a more compact results view. These plots can be selected to analyze the stability of CV and the residual behavior. The CV fit variability plot is available only when cross-validation is performed. It is placed below the goodness-of-fit table for the corresponding fitted dataset.

5 Outputs

The IVIVC app generates interactive results and downloadable reports that summarize fitted model behavior, model-comparison metrics, diagnostic information, and user-selected analysis settings. Output availability depends on the selected approaches, whether fitting succeeds, and whether a prediction dataset was supplied.

Goodness of fit describes agreement with the data used for fitting. Cross-validation checks predictions against observations set aside from that dataset, giving an approximate indication of how well the fitted models may predict similar new observations. Independent (external) validation provides a more direct test of the resulting IVIVC using additional paired in vitro and in vivo experimental data that were not used to develop or select the relationship; an in vitro prediction dataset alone does not provide this independent comparison. Prediction intervals summarize approximate uncertainty under the fitted model and its error assumptions.

Output	Description
Preprocessing outputs	Processed data tables and plots showing the data used for fitting and prediction.
Model fitting outputs	Parameter estimates reported as estimate ± 95% confidence interval where available, fitted plots with error bars, optional CV fit-variability and residual diagnostic plots, and selected performance metrics.
Model comparison outputs	Tables comparing selected approaches, models, metrics, relative model evidence, and cross-validation summaries.
Prediction outputs	Prediction plots and downloadable tables when a prediction dataset is supplied. Depending on the approach and prediction method, the output is either a predicted in vivo time or a predicted in vivo response value.
Selected model & summary of outputs	User-selected report summaries and optional report export for incorporation into external documentation.

5.1 Relative model evidence

Relative model evidence tables compare candidate parametric models using information criteria (i.e., AIC, AIC_c, and/or BIC), showing which models have more support from the observed data after balancing fit quality against model complexity. These comparisons are only meaningful among models fitted to the same data, with the same preprocessing settings, and within the same IVIVC approach.

These tables are shown only when one or more information criteria are selected as metrics. Each selected criterion is interpreted separately because AIC, AIC_c, and BIC can occasionally rank the same models differently. See Appendix A for interpretation guidance, rules of thumb, and limitations.

The relative model evidence tables contain the following quantities.

Table output	What it means
n	Number of observations used to fit the model.
k	Number of fitted model parameters. Models with more parameters can fit more flexible shapes, so information criteria include a complexity penalty.
AIC, AIC_c, or BIC	Information-criterion value for the fitted model. Lower values indicate stronger relative support within the candidate set.
Delta	Difference between a model's criterion value and the lowest value in the candidate set. Values near 0 indicate similar support to the leading model; larger values indicate less support.
Weight	Normalized relative-support score across the candidate set. Weights are comparative summaries, not absolute probabilities that a model is correct.
Evidence ratio	Ratio of the highest weight in the candidate set to the model's weight. Larger values indicate less relative support compared with the leading model.
Support badge	Rule-of-thumb label based on the delta value. Intended to aid user review, not to recommend a model.

5.2 Prediction outputs

The optional prediction dataset contains new in vitro time and response values. After the selected preprocessing is applied, the app uses those values with the relationship established from the fitting dataset. Different prediction methods use the prediction time, the prediction response, or both, and they do not all predict the same quantity. Some methods predict the corresponding **in vivo time**, while others predict the corresponding **in vivo response value**.

The table below summarizes what each prediction method uses and how the result is calculated. These methods answer different prediction questions and should not be treated as interchangeable outputs.

Approach / method	Prediction data used	How the prediction is made	Output
Approach 1: Direct correlation	In vitro time	Applies the fitted relationship between in vitro time and in vivo time: $t_{\text{vivo,pred}} = f(t_{\text{vitro,pred}})$. The prediction response value is not used in this calculation.	In vivo time
Approach 2: Response-value ratio	In vitro time and response	Evaluates the fitted in vitro and in vivo curves at the prediction time, calculates their response ratio, and uses that ratio to scale the observed prediction response: $y_{\text{vivo,pred}} = y_{\text{vitro,pred}}\, f_{\text{vivo}}(t)/f_{\text{vitro}}(t)$.	In vivo response
Approach 2: Time-constant ratio	In vitro time	Scales the prediction time by the ratio of the fitted in vivo and in vitro time constants: $t_{\text{vivo,pred}} = t_{\text{vitro,pred}}(\tau_{\text{vivo}}/\tau_{\text{vitro}})$. The prediction response value is not used in this calculation.	In vivo time
Approach 2: Observed-response inverse mapping	In vitro response	Finds the time at which the fitted in vivo curve reaches the observed prediction response: $t_{\text{vivo,pred}} = f^{-1}_{\text{vivo}}(y_{\text{vitro,pred}})$. The prediction time is not used to calculate the mapped time.	In vivo time

Approach / method	Prediction data used	How the prediction is made	Output
Approach 3: Response-value scaling	In vitro time and response	Uses the fitting datasets to estimate the in vivo/in vitro response ratio at the prediction time, then multiplies the observed prediction response by that ratio.	In vivo response
Approach 3: Time scaling	In vitro time and response	Uses the fitting datasets to estimate the in vivo/in vitro time ratio associated with the prediction response, then multiplies the prediction time by that ratio.	In vivo time

Interpolation estimates between calibration observations; extrapolation extends a relationship beyond the range used to establish it. The available prediction range depends on the method.

Approach 1 can evaluate its fitted time relationship beyond the paired time range used for fitting. Approach 2 response-value-ratio predictions evaluate both fitted curves at the prediction time, which can fall outside either dataset's observed time range. Time-constant-ratio scaling does not restrict input or predicted times to the observed ranges. The app does not automatically identify these results as extrapolations; they require additional caution.

Approach 2 observed-response inverse mapping restricts predicted times to the observed in vivo calibration time range and requires the fitted curve to be one-to-one over that range. Responses outside the observed in vivo calibration response range are reported as unavailable. Approach 3 interpolates scaling factors within the supported calibration time or response range. These lookup limits do not restrict the resulting scaled in vivo response or time to the observed in vivo range.

For Approach 2, response-value-ratio predictions can become unstable when the fitted in vitro response in the denominator is close to zero. Time-constant-ratio scaling uses one overall time scale for each fitted curve, so it is most useful when the in vitro and in vivo profiles have similar shapes and mainly differ in timing. It may not capture differences that change over the course of degradation, such as multiple degradation phases.

Where supported, prediction plots include error bars or intervals that summarize uncertainty under the selected model and method. Narrow intervals do not establish that extrapolated predictions are reliable. For observed-response inverse mapping, interval bounds are restricted to the calibration time range by construction. Their calculation and limitations are described in Section 5.3.

5.3 Uncertainty and error bars

These displays show how much fitted curves or predictions vary based on the fitted model and residual error. They describe approximate model-based uncertainty and are not intended to define acceptability thresholds for regulatory submissions. Applicable credibility and evidence considerations depend on the intended use and applicable FDA guidance [1, 2].

Very large uncertainty bands can extend beyond the initial plot range and remain out of view until the plot is rescaled. When this occurs, the app displays an informational badge near the affected fitting plot. A separate warning is shown if one or more uncertainty bounds are nonfinite or the uncertainty band cannot be calculated. The badge describes the display condition but does not identify its cause; the fitted parameter diagnostics, model form, and available data provide context for interpreting it.

The following descriptions summarize how fitted-parameter uncertainty, fitting-plot bands, and prediction error bars are calculated or interpreted for each approach.

5.3.1 Approach 1: Direct Correlation

Fitting uncertainty

Parameter estimates: fitted parameters are reported as estimate $\pm$ 95% confidence interval. The plus-minus value is calculated from the fitted model's covariance matrix, parameter standard errors, and residual degrees of freedom. It summarizes how precisely each parameter was estimated from the available data.

Fitting plot bands: shaded 95% prediction bands are generated by Monte Carlo sampling of the fitted parameter covariance matrix. The sampled model variance is combined with the residual variance from the fitted data, then converted to a two-sided interval using the appropriate Student's t multiplier when residual degrees of freedom are available.

These bands describe approximate prediction uncertainty around the fitted relationship. They are not intended to define acceptability thresholds; interpretation depends on the fitted model form, local covariance approximation, residual behavior, preprocessing choices, and intended use.

Prediction uncertainty

Prediction plot error bars: when a prediction dataset is supplied, predicted in vivo times are plotted with horizontal error bars. The bars are calculated from the same 95% prediction-band calculation used for the fitted correlation model; the upper and lower bar lengths are the distances from the band mean to the corresponding upper and lower interval bounds at each input prediction time.

The error bars therefore include propagated fitted-parameter uncertainty and residual/model error from the Approach 1 correlation fit.

5.3.2 Approach 2: Independent Model Fitting

Fitting uncertainty

Parameter estimates: each in vitro and in vivo model parameter is reported as estimate $\pm$ 95% confidence interval. The plus-minus value is calculated from the corresponding fitted model's covariance matrix, parameter standard errors, and residual degrees of freedom. It summarizes how precisely each parameter was estimated from the available data.

Time constants: tau values, when available, are calculated from the fitted model and parameter covariance. Reported tau values are shown as tau $\pm$ 95% confidence interval. The interval accounts for uncertainty in the fitted parameters used to calculate the time constant.

Fitting plot bands: the in vitro and in vivo shaded 95% prediction bands are generated separately. For each fitted model, the app Monte Carlo samples the fitted parameter covariance matrix, combines sampled model variance with residual variance, and converts the resulting standard uncertainty to a two-sided interval using the appropriate Student's t multiplier when residual degrees of freedom are available.

Prediction uncertainty

Response-value-ratio prediction error bars: uncertainty is propagated from both fitted models. The app samples each model's parameter covariance matrix, calculates the in vivo/in vitro response-value ratio for each sample, combines variance in that ratio with residual-based ratio uncertainty, and converts that uncertainty to a two-sided 95% interval. The plotted vertical error bars are scaled from the resulting ratio interval at each prediction time.

Time-constant-ratio prediction error bars: when both tau values are valid, uncertainty from both tau values is carried through the ratio calculation. The resulting uncertainty in the scaled prediction time is converted to a symmetric 95% interval, where possible, and plotted as a horizontal error bar.

Observed-response inverse mapping: each preprocessed observed prediction response is mapped to the time at which the fitted in vivo calibration curve reaches that response. This method uses the new in vitro response values directly. It is restricted to the observed in vivo calibration time range and requires the fitted in vivo curve to be one-to-one over that range. Response values outside the observed in vivo calibration response range are not extrapolated. If a response is within the observed range but lies just beyond the fitted curve at a calibration endpoint, it is mapped to that endpoint rather than discarded. Approximate 95% inverse-prediction intervals are generated by Monte Carlo sampling of the fitted in vivo parameter covariance and by treating the in vivo fit's residual standard error as the assumed uncertainty of the mapped response. This residual-scale assumption is a surrogate because the app does not have a separately measured error model for the new in vitro response.

Response-value ratio: predictions and error bars can become unstable when the fitted in vitro response in the denominator is close to zero, because small changes in the fitted curves can then produce large changes in the ratio.

Time-constant ratio: uncertainty can become large if the fitted in vitro time constant is small or poorly estimated. The method also reduces each fitted curve to one overall time scale, so it may not capture differences in curve shape

that change over time.

5.3.3 Approach 3: Direct Mapping

Fitting uncertainty

Parameter uncertainty: Approach 3 does not fit a parametric model, so there is no fitted-parameter covariance matrix, no parameter confidence intervals, and no model-based 95% prediction band.

Fitting plots: the plots show interpolation-based ratios/mappings calculated from the preprocessed datasets. They represent deterministic mappings from the supplied data and preprocessing choices, not statistical uncertainty bands.

Prediction uncertainty

Prediction plot error bars: Approach 3 does not generate formal propagated error bars for prediction outputs because no parametric covariance model is fitted.

Uncertainty is therefore limited to the quality of the input data, preprocessing, interpolation, and the assumption that the observed mapping remains valid for the prediction dataset. Extrapolated points outside the interpolation range are excluded or returned as unavailable.

6 Miscellaneous

6.1 Software verification and validation

Software verification concerns whether the app works as described and performs its calculations correctly. Software validation concerns whether it supports its intended empirical correlation tasks. Sponsors using the app as provided do not need to verify or validate its software functionality. These questions are separate from assessing whether an individual IVIVC is suitable for its intended use.

6.2 Command-line interface

The repository includes `cli.py` for batch-oriented fitting from the command line. Use `python cli.py --help` to see available arguments for input files, approaches, models, metrics, preprocessing options, cross-validation settings, and grid-search settings. The internal keys specified in this user guide are used to specify settings with `cli.py`.

6.3 Extending the app

The IVIVC app is designed so that common extensions can be added with minimal changes to the main application code. In many cases, new models, performance metrics, and preprocessing methods can be added by following the structure of the existing files in the relevant subdirectories.

6.4 Data handling

The app does not save uploaded datasets or fitting results to persistent storage. Uploaded data are processed for the current analysis, and generated results and downloads are returned directly to the user. When running the app locally, no data leaves the user's computer.

Appendices

A How to Interpret Model Performance

No single metric determines whether an IVIVC model is meaningful. An evaluation of model performance can include goodness of fit, prediction error, model complexity, cross-validation behavior, residual patterns, parameter uncertainty, and whether the fitted relationship is scientifically plausible for the material, degradation mechanism, and intended use. Performance metrics are descriptive and comparative tools, not acceptance criteria. Small datasets can produce unstable estimates, and similar metric values do not necessarily indicate equivalent scientific validity. For analyses intended to support regulatory submissions, the depth of supporting evidence depends on the model's intended use and applicable credibility considerations [1, 2].

The app reports metrics and comparison tables as descriptive aids for user review; the scope and advisory status described in the Introduction apply throughout this appendix. Users are responsible for determining how to document preprocessing assumptions, fitting settings, model-selection rationale, limitations, and any planned use of the model outside the fitted data range, consistent with the intended use and applicable credibility considerations.

A.1 Model comparability before interpreting metrics

Performance metrics are most meaningful when the models being compared were evaluated under identical conditions. Metric comparisons are most interpretable within the same uploaded dataset, preprocessing configuration, response scale, selected IVIVC approach, and model set.

In particular, AIC, AIC_c, and BIC values are not comparable across different datasets, transformed responses, preprocessing settings, or substantially different fitting contexts. These information criteria measure the *relative* support among models evaluated under the same conditions—they are not absolute quality scores and lose their meaning when the underlying data or setup differs. Approach 3 is interpreted differently from Approaches 1 and 2 because it uses direct interpolation-based mappings rather than fitting a parametric model.

A.2 Metric ranges, direction, and interpretation

Metric	Range / direction	Interpretation guidance
R^2	Best possible value is 1. Values near 0 indicate little improvement over predicting the mean response. Can be negative if the model performs worse than the mean-response baseline. Higher is better.	Proportion of response variance explained by the model [13]. High R^2 does not establish that the model is mechanistic, causal, predictive outside the fitted data range, or adequately supported for the intended use.
Adjusted R^2	Higher is better. Can be lower than R^2.	Penalizes additional fitted parameters, making it more informative than R^2 when comparing models of different complexity [13].
MAE	Lower is better. Units match the response units.	Mean absolute error on the response scale [11]. Interpret relative to assay variability, measurement uncertainty, and the intended use.
MSE	Lower is better. Units are squared response units.	Mean squared prediction error [11]. Useful computationally, but less intuitive than RMSE because the units are squared.
RMSE	Lower is better. Units match the response units.	Root mean squared error, the typical prediction error on the response scale [11]. Interpret relative to assay variability, measurement uncertainty, and the intended use. Penalizes larger errors more heavily than MAE.
NRMSE	Lower is better. Normalized to allow more meaningful comparison across response scales.	RMSE prediction error expressed relative to a normalization factor such as the response mean or standard deviation. Interpretation depends on the normalization method used.

Metric	Range / direction	Interpretation guidance
AIC	Lower is better, but only relative to other comparable models. No fixed bounds; negative values are allowed.	Balances goodness of fit against model complexity [19, 14, 20]. A lower value indicates a model that explains the data well without unnecessary parameters. Absolute values are not standalone quality grades.
AIC_c	Lower is better, but only relative to other comparable models. Undefined when there are too few observations relative to the number of fitted parameters.	A small-sample correction to AIC that is most relevant when the observation-to-parameter ratio is low [21, 14, 20]; $n/k < 40$ is a rule of thumb [14].
BIC	Lower is better, but only relative to other comparable models. No fixed bounds; negative values are allowed.	Balances goodness of fit against model complexity, typically applying a stronger penalty for additional parameters than AIC [22, 14, 20].
Cross-validation metrics	Direction follows the underlying metric: higher is better for R^2-type metrics, lower is better for error metrics.	Estimate prediction performance under the selected splitting scheme [15, 16, 17, 18]. Can help identify overfitting, instability, or sensitivity to individual observations or data splits.
Fit/CV ratio	Values near 1 indicate similar fitted and cross-validation performance.	A ratio near 1 suggests similar full-fit and cross-validation behavior. Substantial deviations can indicate overfitting, instability, or sensitivity to the splitting scheme.

A.3 Interpreting goodness of fit and prediction error

Goodness-of-fit metrics—R^2, adjusted R^2, MSE, MAE, RMSE, and NRMSE—describe how well the model fits the data used to estimate its parameters. Their interpretation depends on practical context, not just numerical value. The app does not apply a universal IVIVC performance cutoff; interpretation depends on the intended use and, for regulatory contexts, applicable FDA guidance [1, 2].

Strong goodness-of-fit can still be misleading, particularly for small datasets or models with many parameters. Favorable fit metrics do not rule out unstable parameters, poor cross-validation performance, systematic residual patterns, or a fitted curve shape that is not scientifically plausible for the degradation process being modeled.

RMSE is often the most intuitive error metric because it is expressed in the same units as the response. NRMSE can be useful for comparing error across different response scales, but the choice of denominator changes what the metric actually measures [12]:

- **NRMSE normalized by the response mean** answers: "How large is the typical prediction error relative to the average response value?" This normalization is useful for comparing model error across endpoints with different scales or units. For example, comparing how well a model fits mass loss data versus molecular weight data on a common scale.
- **NRMSE normalized by the response standard deviation** answers: "How large is the typical prediction error relative to the natural spread of the observed responses?" This normalization is useful for gauging how much of the variability in the data the model fails to capture. A value approaching or exceeding 1 indicates that the typical prediction error is as large as the spread of the data itself, suggesting the model explains little of the observed variability, analogous to an R^2 near zero.

NRMSE values are not directly comparable across datasets or response scales when the normalization method differs. More broadly, the practical importance of a given error level depends on context: a numerically small error may still matter when the relevant degradation endpoint has narrow variation, while a larger error may be less consequential during exploratory model development when high predictive precision is not central to the analysis.

A.4 Interpreting relative model evidence

Relative model evidence tables compare candidate models using information criteria (i.e., AIC, AIC_c, and BIC), which balance goodness of fit against model complexity. In this context, *empirical support* means that the observed data provide more support for one model than another within the displayed candidate set, based on the selected information criterion. It does not establish that a model is correct, mechanistic, or predictive outside the observed range. Whether the evidence base is sufficient to support a regulatory submission depends on the intended use and applicable FDA guidance [1, 2].

Rules of thumb for delta values

Delta is the difference between a model's criterion value and the lowest value in the candidate set. Common interpretive conventions are:

- **AIC / AIC_c**: delta 0–2 suggests similar support; 4–7 suggests less support; greater than 10 suggests essentially no support relative to the leading model [14].
- **BIC**: delta 0–2 suggests similar support; 2–6 suggests less support; 6–10 suggests substantially less support; greater than 10 suggests essentially no support relative to the leading model [23].

These are interpretive conventions, not regulatory acceptance criteria.

Evidence weights

For a given criterion, the evidence weight for model i is:

$$w_i = \frac{\exp\left(-\frac{1}{2}\Delta_i\right)}{\sum_{j=1}^{J} \exp\left(-\frac{1}{2}\Delta_j\right)} \tag{1}$$

where Δ_i is the delta value for model i and J is the number of candidate models. Larger weights indicate stronger relative support within the candidate set [14, 20].

When multiple models have similar support

If several models have similar delta values or weights, additional considerations can include model simplicity, parameter plausibility, residual behavior, cross-validation performance, and scientific interpretability.

A.5 Interpreting cross-validation performance

Cross-validation evaluates prediction performance using observations left out when the model was fit. In this app, it is most useful as a **sensitivity diagnostic**: when the cross-validation error is similar to the error from fitting the full dataset, this suggests the fitted relationship is reasonably stable regardless of which observations are set aside. When the two differ substantially, the model may be more sensitive to which specific data points were included in fitting, or to the particular way the data were split. Cross-validation does not measure uncertainty in the model parameters. The practical question to keep in mind is not just "how good is the score?" but also "what kind of prediction task did the cross-validation scheme actually create?"

What each scheme is testing

Different cross-validation schemes answer different questions, and those questions determine how the results are interpreted. Shuffle split, the default, sets aside randomly selected time points without regard to their order. It serves as a general check of overall curve-shape robustness, but it may make the model appear more reliable than it is when nearby time points carry similar information. Leave-one-timepoint-out tests how well the model predicts a single omitted time point when all others are available, a relatively gentle task when neighboring time points are close together. Leave-contiguous-block-out tests whether the model can bridge an entire missing time window, a harder and more time-aware task that is more sensitive to gaps in the profile.

These distinctions matter because adjacent time points in a degradation profile are often similar to one another, measurement errors may be related across time, and early, middle, and late portions of a degradation profile may reflect

qualitatively different behavior. Standard cross-validation procedures are reviewed in [15]; additional considerations for temporally ordered or dependent data are discussed in [16, 17, 18].

Interpreting results for small IVIVC datasets

For the small datasets typical in IVIVC development, cross-validation results are intended as an exploratory diagnostic or sensitivity analysis rather than a definitive performance benchmark. Interpretation can include comparison of the CV metric with the corresponding metric for the model fitted to the full dataset together with whether the chosen splitting pattern reflects the practical prediction question of interest.

For example, strong leave-one-timepoint-out performance alongside weaker leave-contiguous-block-out performance may indicate that the model can interpolate individual missing points well, but becomes less reliable when an entire local region of the profile is absent from fitting. The importance of that distinction depends on the intended use of the model. For credibility assessment in the context of regulatory submissions, see FDA guidance on computational modeling credibility and reporting [1, 2].

CV fit variability plots

Average CV performance metrics summarize prediction performance across the held-out observations and CV splits, but they do not show how much the fitted curve itself changes from one training subset to another. Two models can therefore have similar average CV errors even when one produces nearly the same fitted relationship across splits and the other changes appreciably depending on which observations are withheld.

The optional CV fit variability plot provides this complementary information by showing the full-data fit together with the mean fitted curve and mean $\pm$ one standard deviation across the successful CV training-set fits. A narrow band indicates that the fitted relationship changes relatively little across the CV splits. A wider band, or widening confined to a particular part of the curve, identifies regions where the fitted relationship is more sensitive to the training data, suggesting that additional data in that region may be particularly informative. A noticeable difference between the mean CV fit and the full-data fit can likewise indicate that repeatedly leaving out observations changes the fitted relationship in a systematic way.

The band is a descriptive sensitivity diagnostic, not a confidence interval, prediction interval, or parameter-uncertainty interval. Its appearance depends on the selected CV scheme and the number of successful CV fits; interpretation therefore also depends on the average CV metrics and the prediction task represented by the chosen splitting scheme.

Fit/CV ratios

The app reports fit/CV ratios to summarize the relationship between full-fit and cross-validation performance for the same metric. A ratio near 1 indicates that the two are similar. A ratio far from 1 may suggest reduced generalizability, overfitting, model instability, or sensitivity to the splitting scheme. Because the direction of concern depends on the metric (for example, a ratio above 1 is concerning for RMSE but not for R^2), interpretation of the ratio depends on the raw fit and CV values, the chosen splitting scheme, and the fitted plots.

Cross-validation comparison badges

The app displays cross-validation comparison badges as interpretive aids. Badges are assigned based on the preferred direction of the metric: for lower-is-better metrics such as RMSE, a worse CV result means the CV value is larger than the full-fit value; for higher-is-better metrics such as R^2, a worse CV result means the CV value is smaller. The thresholds used are:

- **Similar**: CV performance is within a factor of 1.25 of the full-fit value.
- **Moderately worse**: CV performance is 1.25- to 2-fold worse than the full-fit value.
- **Substantially worse**: CV performance is more than 2-fold worse than the full-fit value.

These thresholds are heuristic review aids to help flag results that warrant closer attention. They are not statistical acceptance criteria or regulatory standards.

Metrics excluded from cross-validation comparisons

The app does not include adjusted R^2, information criteria (AIC, AIC_c, and BIC), or normalized RMSE (NRMSE) in cross-validation comparison tables. The reason is that these metrics are not directly comparable across fits that use different numbers of observations:

- **Adjusted R^2 and information criteria** incorporate the number of observations or fitted parameters as part of their calculation. Because the full-data model and each cross-validation model are fitted to different subsets of the data, these quantities are not on the same scale and are not directly comparable.
- **NRMSE** is normalized by a factor that generally differs across subsets. This means the full-fit and CV values can have different denominators, making a direct comparison misleading.

Eligible cross-validation metrics summarize prediction behavior during CV, while the ineligible CV metrics remain available for evaluating and comparing fitted models. Not all eligible metrics are available for every cross-validation scheme because some require more than one observation to be defined or stable. For example, R^2 cannot be calculated for a single omitted point and will be unavailable under leave-one-timepoint-out CV; interpretation then depends on the metrics that are reported.

A.6 Interpreting residual diagnostics

Residuals are the differences between observed and predicted values (i.e., the errors in the predictions). A well-specified model generally produces residuals without systematic structure. Residual diagnostics help assess whether model errors show systematic patterns [24]. The app provides two qualitative diagnostic plots: residual-vs-time plots and quantile-quantile (Q-Q) plots. If your dataset contains few time points, which is typical of IVIVC datasets, these plots are intended as exploratory diagnostics rather than pass/fail tests. With only a few points, random variation can sometimes appear to form a meaningful pattern.

Residual-vs-time plots

Residuals are plotted in the same units as the fitted response. The axis covers the largest absolute residual for each model but is not allowed to shrink below a small fraction of the observed response range. This minimum prevents extremely small numerical residuals from being expanded to fill the plot and appearing more important than they are. Overall error metrics such as RMSE or MAE provide a more direct comparison of error magnitude across models than residual plots.

Q-Q plots

Q-Q plots compare the ordered standardized residuals against the pattern expected for randomly distributed errors (i.e., a normal distribution, Gaussian distribution, or 'bell curve'). Points near the diagonal reference line are broadly consistent with random residual variation. Systematic curvature, large departures in the tails, or isolated extreme points may suggest non-randomness, outliers, or a model form that does not fit the data well. When residual variation is too small relative to the response scale, the app reports that the Q-Q plot is not informative rather than magnifying numerical differences.

What to look for

Residual diagnostics are intended for qualitative review—by eye—rather than use as formal statistical tests. Quantitative tests are available for problems such as non-random errors, correlated errors across time (autocorrelation), and variance that changes across the response range (heteroscedasticity) [24]. However, with the small numbers of observations typical of IVIVC datasets, these tests may have limited ability to distinguish systematic problems from random variation. Visual patterns are prompts for further review rather than definitive findings. Residuals scattered around zero in the residual-vs-time plot, and points near the diagonal in the Q-Q plot, are consistent with a model form that captures the observed pattern, though they do not establish model validity. The following patterns may indicate bias or a model form that does not fully capture the data:

- Sustained curvature or systematic trends in the residuals over time.
- Clustering, where residuals at adjacent time points are consistently on the same side of zero.

When any of these patterns appear, they can motivate evaluation of whether a different model form or a transformed (i.e., normalized or scaled) response better describes the underlying relationship. For a visual example and additional references for interpreting residual plots, see Appendix B.

A.7 Interpreting parameter uncertainty

Model parameters cannot always be estimated precisely from the available data. The app reports each parameter estimate ± 95% confidence interval. The plus-minus value represents the confidence interval half-width around the estimate. In practical terms, a wide interval means the data do not pin down that parameter precisely, while a narrow interval means the estimate is more stable.

The interval is calculated from the fitted model's covariance matrix and residual degrees of freedom. This calculation estimates uncertainty from the behavior of the model near the fitted parameter values. For nonlinear models, especially with small datasets, the true uncertainty can be wider, asymmetric, or more strongly affected by relationships among parameters than the reported interval suggests. The interval is most informative when considered together with the relative standard errors, parameter-coupling diagnostics, fitted curves, and scientific plausibility rather than as a precise boundary for the parameter. For a practical overview of nonlinear regression and interpretation of fitted results, see [25]. For a technical treatment of covariance-based parameter uncertainty and approximate confidence intervals in nonlinear least-squares models, see [26].

The app summarizes parameter uncertainty using informational badges as screening aids, not acceptance criteria. They highlight fitted-parameter uncertainty that may affect interpretation or prediction.

Diagnostic	**What it summarizes**	**Interpretation guidance**
Relative standard error	The uncertainty in a parameter estimate expressed as a percentage of the estimate itself.	A large value means the uncertainty is large relative to the estimate, so the data are not pinning down that parameter well. The app flags values of at least 50% as elevated and at least 100% as large.
Parameter coupling / correlation	How strongly pairs of parameter estimates move together during fitting (pairwise correlation).	When two parameters are highly correlated, the model can reach a similar fit by trading off one against the other, making both harder to estimate reliably. Some coupling is normal in nonlinear models. The app reports correlation alone as *parameter coupling noted*; when high correlation coincides with elevated relative standard error or a numerical sensitivity flag, it reports a stronger identifiability caution. Absolute correlations of at least 0.90 are noted, with stronger concern when correlations of at least 0.95 occur alongside other uncertainty diagnostics.

If no parameter uncertainty concerns are flagged, the app did not detect large relative standard errors or strong parameter coupling under these screening checks. This does not establish mechanistic correctness, predictive adequacy for a particular use, or regulatory acceptability. Conversely, a coupling or warning badge does not by itself determine whether the model can support a particular use; rather, it identifies parameter behavior that can be reviewed alongside the fitted curves, residuals, cross-validation behavior, and scientific plausibility.

Parameter uncertainty deserves particular attention when a model will be interpreted mechanistically, used to extrapolate beyond the observed data range, or used to compare degradation behavior across materials, processes,

or test conditions. In these situations, relevant considerations include whether the reported intervals are wide, parameters are strongly coupled, fitted parameter values are extreme or scientifically implausible, or small changes in the data or fitting settings produce substantially different estimates. Strong fit metrics alone may not be sufficient to address concerns about poorly estimated parameters or scientifically implausible values; these aspects warrant separate review.

A.8 Considering model complexity

More complex models (i.e., those with more parameters) may fit observed data more closely, but with small datasets this may be a liability rather than an asset. A flexible model may trace the noise in the data rather than the underlying pattern, and its parameters may be poorly estimated as a result, an issue called *overfitting*. Adjusted R^2, AIC, AIC_c, BIC, cross-validation metrics, and residual diagnostics provide complementary information about whether added complexity is justified [27].

When two models have similar fit and similar empirical support, model simplicity and scientific plausibility are among the factors that may support the rationale for the selected model.

A.9 Scientific plausibility and intended use

Statistical performance alone is not sufficient to evaluate a model. Scientific plausibility also depends on what is known about the material, degradation mechanism, and test conditions. A model that is statistically competitive but produces a fitted shape, parameter values, or extrapolated behavior that conflicts with known degradation behavior may warrant additional rationale regardless of its metrics.

The intended use of the model—including whether it will inform a regulatory submission—affects the type and depth of supporting evidence that may be relevant. Applicable considerations are described in FDA guidance on computational modeling credibility and reporting [1, 2].

A.10 Example model review workflow

The following is one possible sequence of review steps for illustrative reference. It does not establish a required procedure or replace applicable credibility and submission guidance [1, 2].

1. Confirm that the dataset, preprocessing choices, and selected response values (the measured values being modeled) are appropriate for the intended analysis.
2. Review fitted curves and residual plots for each candidate model, checking for systematic patterns or implausible fitted shapes.
3. Compare goodness-of-fit metrics such as R^2 and RMSE in the context of assay variability, endpoint importance, and intended use.
4. Review cross-validation performance and fit/CV ratios to assess sensitivity to the available data.
5. Review relative model evidence using information criteria to compare fit quality against model complexity.
6. Review fitted-parameter estimates, 95% confidence intervals, uncertainty diagnostics, and scientific plausibility.
7. Consider whether model complexity is justified by the available data and the degradation mechanism.
8. Document the rationale for the user-selected model based on a balance of statistical performance, scientific plausibility, and intended use.

A.11 Questions to guide model review

The following questions illustrate considerations that a user might explore. They do not define a required review process or imply that each question must be addressed in a regulatory submission.

1. **Does the model describe the observed data well?**
    - Does the fitted curve follow the main trend in the data?

- How do performance metrics such as R^2, RMSE, and NRMSE compare in the context of the intended use?
- How large is the prediction error relative to assay variability, measurement uncertainty, or the degradation endpoint being evaluated?
- Are there regions where the model consistently overpredicts or underpredicts?

2. **Does the model generalize beyond the fitted data?**
   - Are cross-validation metrics similar to the metrics for the model fitted to the full dataset, and are fit/CV ratios near 1?
   - Could the cross-validation result be sensitive to the splitting scheme, number of observations, or a small number of influential points?
   - If cross-validation performance is poor, how does that affect interpretation of the model for its intended purpose, including whether a simpler or more stable form may warrant consideration?
3. **Are the model parameters interpretable and stable?**
   - Are the fitted parameter values scientifically plausible?
   - What do the 95% confidence intervals indicate about the precision of the parameter estimates?
   - Are any parameters strongly coupled—meaning different combinations of parameter values could produce a similar fitted curve?
   - If coupling is present, does it coincide with large parameter uncertainty or an identifiability caution?
   - Are individual parameter values being interpreted mechanistically, or is the model being used mainly to describe the overall curve shape?

   A model can have a fitted curve that closely follows the observed data while still having uncertain or coupled parameters. In that case, the curve may be useful for empirical description within the observed data range. Interpret individual parameter values cautiously.
4. **Is the model unnecessarily complex?**
   - Does the more complex model improve fit, cross-validation behavior, or relative model evidence enough to support the added parameters?
   - Do AIC, AIC_c, or BIC provide support for the additional complexity within the candidate model set?
   - If two models perform similarly, how do model simplicity and scientific plausibility contribute to the rationale for the selected model?
   - Does the dataset contain enough observations to inform the fitted parameters?
5. **Is the model scientifically plausible?**
   - Is the fitted curve shape consistent with expected material degradation behavior?
   - How do the model assumptions align with the material, test conditions, and degradation endpoint?
   - How does the model behave near the edges of the observed data range, and how would that behavior affect interpolation or extrapolation?

   A statistically competitive model may still warrant additional rationale if its shape, parameter values, or extrapolated behavior conflict with the known degradation process.
6. **How does the intended use affect interpretation of the model?**
   - Is the model being used for exploratory analysis, interpolation, prediction, comparison across conditions, or support for a regulatory rationale?
   - How does the level of uncertainty affect interpretation for that purpose?
   - Are the model's limitations clearly documented?

- Would an independent reviewer understand the rationale for selecting this model over other candidate models?

Potential elements of a model-selection rationale include fitted performance, cross-validation behavior, relative empirical support, parameter diagnostics, residual behavior, scientific plausibility, and intended use.

A.12 Example interpretation statements

The following passages illustrate how model results might be described in a technical context. They are examples only and are not FDA-endorsed language for regulatory submissions. Adapt all language to the specific analysis and supporting evidence.

- Model A had the lowest RMSE and highest adjusted R^2 among the evaluated models, indicating the strongest goodness-of-fit under the selected preprocessing conditions.
- Model A was selected over the more complex Model B because both models showed similar fit and AIC_c support (Delta $AIC_c < 2$), so the observed data provided little additional support for the added complexity of Model B.
- Although Model C had strong goodness-of-fit, its cross-validation error was substantially higher than its fitted error, suggesting sensitivity to which observations were included in fitting.
- Residuals for Model A were scattered around zero without obvious systematic patterns, consistent with a model form that captures the observed data.
- Model B showed a systematic trend in the residuals across time, suggesting the selected model form does not fully capture the degradation profile.
- The fitted parameters for Model A were within plausible ranges for the degradation mechanism, with narrow confidence intervals indicating stable estimation.
- Model C showed strong parameter coupling between the rate and shape parameters, with relative standard errors exceeding 100%. Parameter estimates were retained for curve description but were not interpreted individually.
- The selected model was chosen based on a balance of goodness-of-fit, cross-validation behavior, relative empirical support, parameter stability, and scientific plausibility for the degradation mechanism and intended use.

B Worked Example: Comparing Candidate Models

B.1 Purpose of the example

This worked example illustrates how fitted curves, performance metrics, cross-validation, residual plots, Q–Q plots, and parameter uncertainty results can be considered together when comparing candidate models. It provides practical examples of residual patterns that can help readers interpret their own results; further illustrated examples are available in [25, 24, 28, 29, 30]. The dataset is simulated so the underlying relationship is known, but the comparison is presented in the same way as an analysis of an unknown experimental relationship. The example is intended as a practical demonstration, not as an acceptance procedure or a set of model-selection criteria.

An accompanying Python script generates a noisy single-exponential decay dataset and fits four candidate models. The four candidate models fitted to the primary dataset are:

- a linear model, included as a form that does not match the data-generating process and is expected to demonstrate clear lack of fit;
- an exponential model, which matches the form used to generate the data;
- a stretched-exponential model, which can reduce error slightly by adding a shape parameter; and
- a double-exponential model, which can reduce error further while adding parameters that may be weakly estimated and strongly coupled.

This example uses the same fitting functions, performance metrics, shuffle-split cross-validation constructor, and relative evidence calculations used by the app. The figures use a simplified layout rather than reproducing the interactive app displays exactly.

B.2 Simulated dataset and analysis settings

The simulated dataset has the form

$$y = a \exp(bt) + \varepsilon,$$

where ε is independent Gaussian noise. The random seed, parameter values, noise level, and cross-validation settings are fixed so the example is reproducible.

Setting	Value
Generating model	Exponential
True parameter a	100.000
True parameter b	-0.075
Noise standard deviation	3.000
Dataset random seed	27
CV scheme	Shuffle split
CV splits	100
CV test fraction	0.25
CV random seed	12345

B.3 Fitted curves

Figure 1 shows the simulated observations with all four fitted curves. The fitted curve display is the first opportunity to identify clearly implausible behavior, but models that differ meaningfully can appear similar when the response scale is large relative to the residual differences. For this example, the linear model visibly misses the decay shape. The three exponential-family models can appear similar at the scale of the response plot, even though the stretched and double-exponential forms use additional parameters.

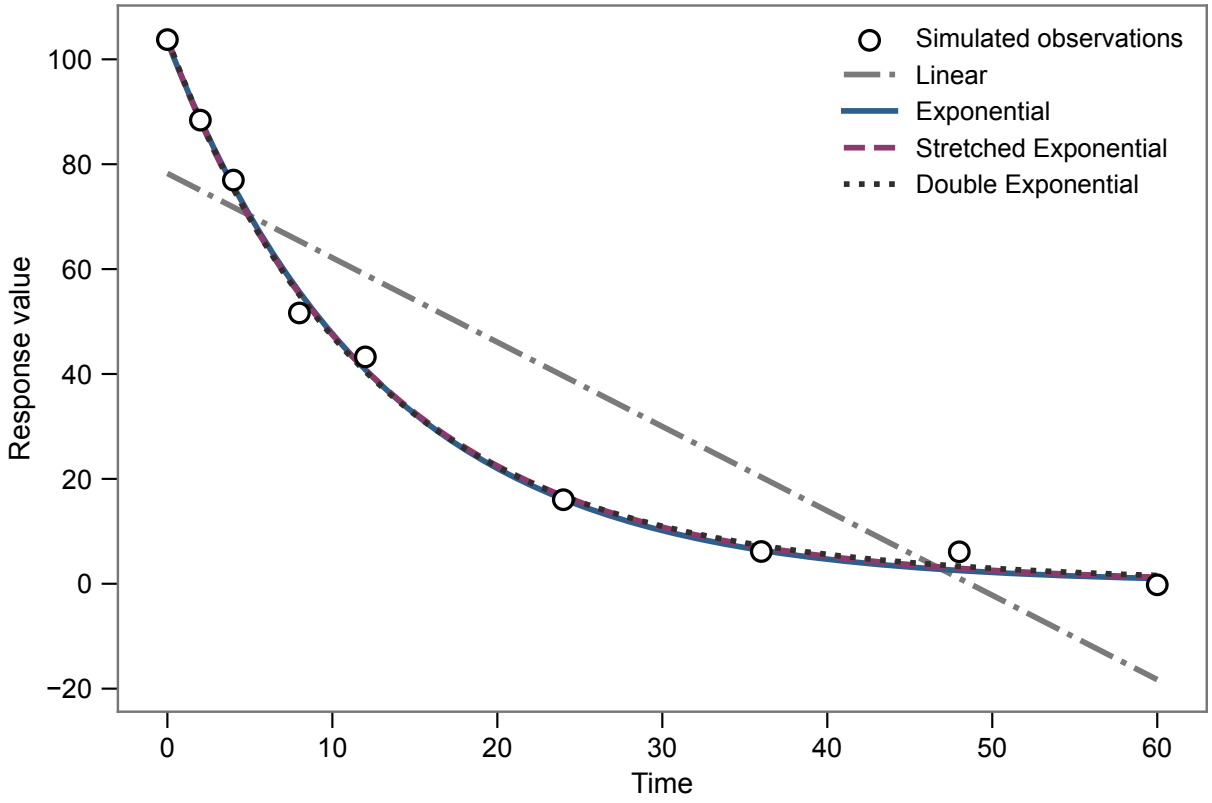


Figure 1: Candidate models fitted to the simulated dataset.

B.4 Fit metrics, model complexity, and cross-validation

Table 17 reports goodness-of-fit metrics, AIC_c comparisons, cross-validation error, and a compact parameter-correlation summary. These quantities answer different questions. R^2 and RMSE describe agreement with the fitted observations. AIC_c compares relative empirical support while penalizing additional fitted parameters. Cross-validation RMSE describes prediction error for omitted observations. The ratio of cross-validation RMSE to full-fit RMSE provides a simple indication of how much prediction error increases when part of the dataset is excluded from fitting.

Table 17: Model performance summary.

Model	R^2	RMSE	AIC_c	Delta AIC_c	CV RMSE	CV/fit
Linear	0.805	16.338	56.283	37.381	24.279	1.486
Exponential	0.997	2.048	18.902	0.000	2.783	1.359
Stretched Exponential	0.997	1.987	23.156	4.254	3.557	1.791
Double Exponential	0.997	1.960	30.112	11.209	1261.292	643.566

In addition to its visible lack of fit, the linear model has substantially poorer fit metrics. Its R^2 remains moderately high because it captures part of the large overall change in response, even though it does not describe the curved decay pattern well. The exponential model provides the strongest overall balance: it represents the generating relationship with only two parameters and has the lowest AIC_c. The stretched and double-exponential models reduce RMSE slightly, but their higher AIC_c values indicate that these small improvements do not offset the added complexity. Cross-validation provides a separate check on how prediction error changes when observations are omitted from fitting. The cross-validation results are similar to the full-fit results for the linear, exponential, and stretched-exponential models. In contrast, the double-exponential model gives much poorer cross-validation performance, indicating greater sensitivity to the observations used for fitting.

B.5 Residuals versus time and missed structure

Residuals are calculated as observed minus predicted response (i.e., the prediction errors). A fitted form that captures the observed pattern generally leaves residuals distributed around zero without sustained curvature or adjacent groups that remain mainly above or below zero. Figure 2 uses the same y-axis range for all four models so the sizes and patterns can be compared directly.

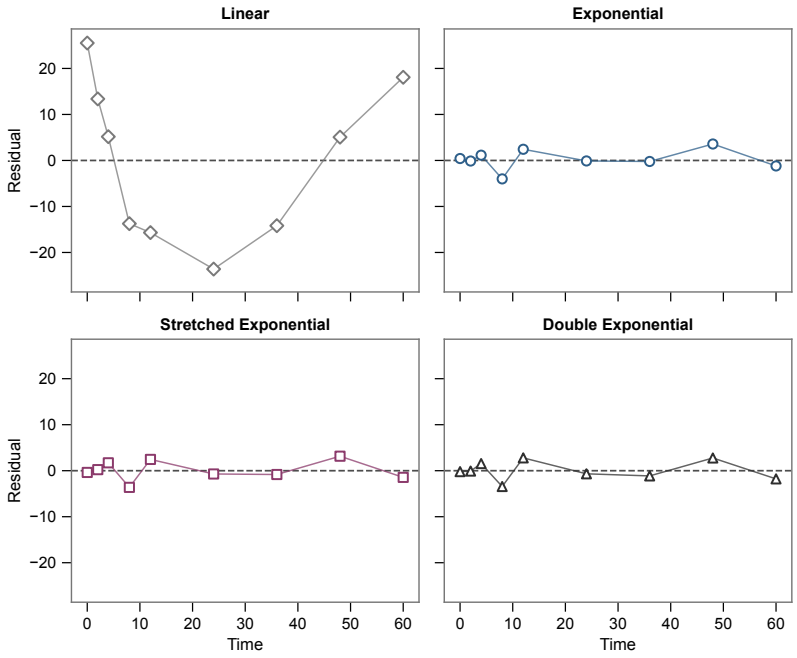


Figure 2: Residuals versus time for the four fitted models.

The linear residuals show a pronounced systematic pattern, indicating that the model does not represent the curved decay well. The exponential residuals fluctuate around zero without any sustained structure. The stretched and double-exponential models produce similarly centered residuals that are slightly smaller at some observations; that modest improvement can be interpreted alongside their added complexity, AIC_c, cross-validation behavior, and parameter uncertainty.

B.6 Q–Q plots and residual distribution shape

A normal Q–Q plot compares ordered standardized residuals with the quantiles expected from a normal distribution. Points close to the reference line are broadly consistent with the assumed residual distribution. Curvature, separation at the ends, or an isolated point can draw attention to skewness, heavy tails, or an unusual residual. With a small dataset, the plot has limited resolution and is interpreted together with the residual-versus-time plot rather than as a formal confirmation of normality.

These Q–Q plots do not distinguish the candidate models as clearly as the residual-versus-time plots, largely due to the small number of observations. A Q–Q plot describes the residual distribution, whereas a residual-versus-time plot is more directly informative about systematic lack of fit across the observed profile. For this dataset, the app would not display Q–Q plots for the exponential-family models because the residual variation is too small relative to the overall response range. Instead, it would report that the Q–Q diagnostic is not informative rather than magnifying very small numerical differences.

B.7 Effect of the number of observations

Figures 1–3 use the primary dataset with 9 observations. To illustrate the effect of denser sampling, a second independent dataset was generated from the same single-exponential relationship and noise distribution at 61 evenly spaced times from 0 to 60. Figure 4 shows the fitted curves and residual diagnostics for the linear and exponential

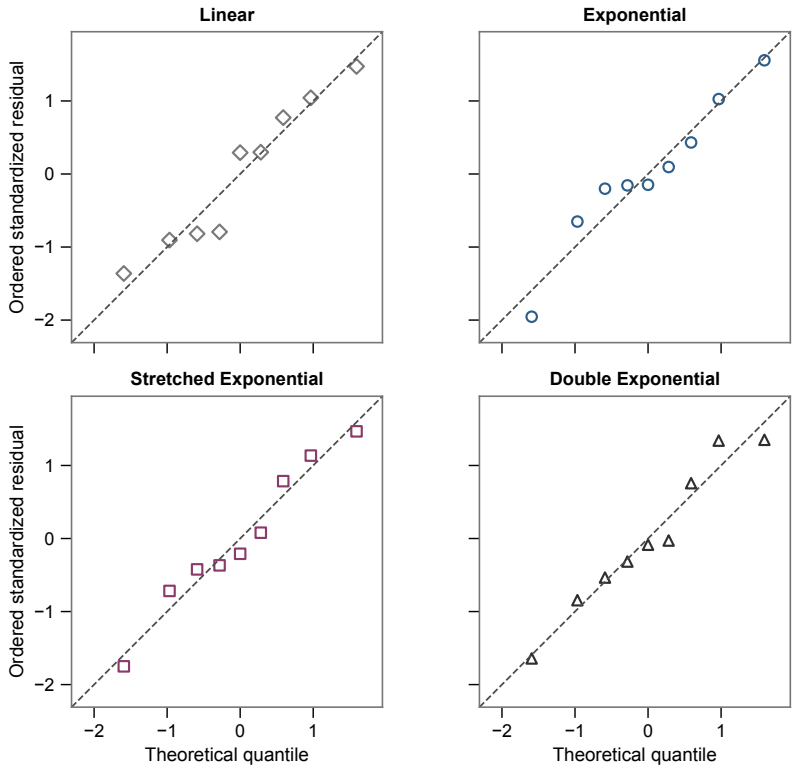


Figure 3: Normal Q–Q plots of standardized residuals.

models. This section provides a qualitative example of how the number of observations can affect the ability to distinguish a model that captures the observed pattern from one with systematic lack of fit.

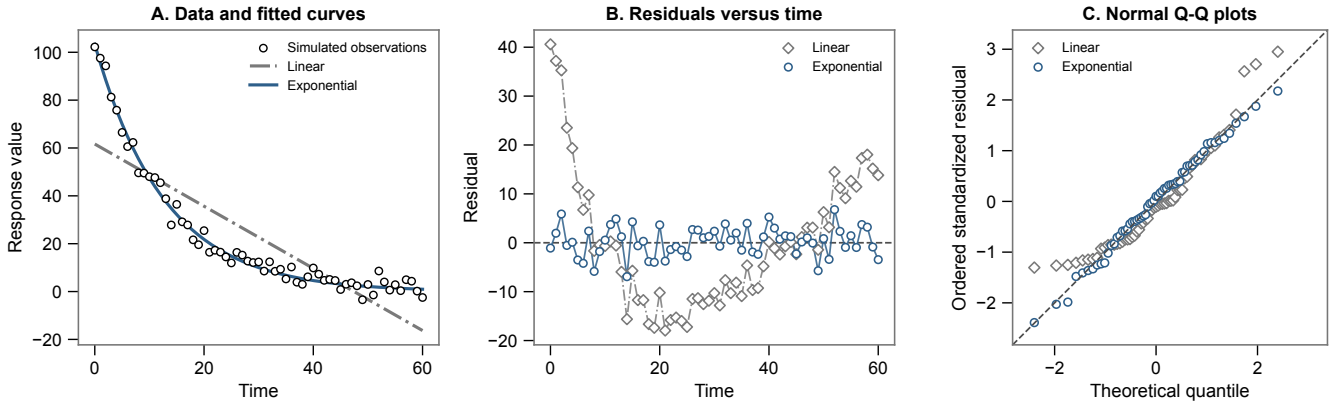


Figure 4: Graphical diagnostics for linear and exponential fits to a simulated dataset with 61 evenly spaced observations: fitted curves, residuals versus time, and normal Q–Q plots.

The smaller dataset already suggests that the linear model does not capture the decay relationship, but the denser dataset makes the lack of fit easier to diagnose. With more observations, the linear residuals form a clearer systematic pattern over time rather than fluctuating randomly around zero, and departures from the Q–Q reference line are easier to assess. In contrast, the exponential-model residuals remain comparatively patternless around zero and follow the Q–Q reference line more closely. Additional observations do not guarantee that a selected model captures the underlying relationship, but they provide more information for distinguishing systematic lack of fit from random variation. With sparse data, the same patterns may be less apparent or more easily mistaken for noise.

B.8 Fitted-parameter uncertainty

The app reports each fitted parameter with a 95% confidence interval. Table 18 also reports the standard error (SE) and relative standard error (RSE = SE / estimate in %).

Table 18: Fitted-parameter estimates and approximate uncertainty.

Model	Parameter	Estimate	SE	95% CI	RSE (%)
Linear	a	78.242	8.936	21.131	11.421
Linear	b	-1.609	0.300	0.709	18.628
Exponential	a	103.344	1.758	4.157	1.701
Exponential	b	-0.077	0.003	0.008	4.257
Stretched Exponential	a	104.141	2.281	5.580	2.190
Stretched Exponential	b	-0.085	0.014	0.034	16.204
Stretched Exponential	c	0.964	0.057	0.139	5.874
Double Exponential	a	64.400	355.320	913.379	551.739
Double Exponential	b	-0.096	0.156	0.401	161.667
Double Exponential	c	39.562	356.112	915.415	900.144
Double Exponential	d	-0.056	0.133	0.343	238.297

Table 19 reports the strongest pairwise parameter correlation for each fitted model. Strong correlation can indicate that different combinations of parameter values produce similar fitted curves, especially in flexible nonlinear models. The app considers correlation together with relative standard error: high correlation alone is not unusual, but high correlation combined with large parameter uncertainty provides stronger evidence that the model parameters cannot be uniquely identified.

Table 19: Largest absolute pairwise parameter correlation for each fitted model.

Model	Maximum absolute correlation	Parameter pair
Linear	0.723	a-b
Exponential	0.604	a-b
Stretched Exponential	0.963	b-c
Double Exponential	1.000	a-c

The exponential parameters are expected to be estimated comparatively precisely because the fitted model matches the generating relationship, and indeed their relative standard errors are only a few percent. For the stretched exponential, the estimated shape parameter remains close to the single-exponential value of 1, so the added parameter changes the fitted curve only slightly and does not improve AIC_c. In the double-exponential fit, both components retain nonzero amplitudes (a and c), but the amplitudes and rate parameters (b and d) have very large uncertainty and are almost perfectly coupled. The fitting procedure can trade one component against the other while producing nearly the same overall curve. In the app, this combination of high parameter correlation and elevated relative standard errors would receive an *identifiability caution* badge. This illustrates why a slightly closer fitted curve does not necessarily support the added model structure or interpretation of the individual component parameters.

B.9 Integrated interpretation

No single output determines the preferred candidate. In this example, the linear model is readily set aside because the fitted curve and residual pattern show clear lack of fit. The exponential model provides the strongest overall balance: it captures the curve shape without obvious residual structure and does so with the fewest parameters among the plausible candidates. The stretched and double-exponential models can achieve slightly lower error, but their higher AIC_c values indicate that the improvement does not justify the added complexity for this dataset. The double-exponential parameter results provide an additional caution because both fitted components remain active while their amplitudes and rates are extremely uncertain and strongly coupled.

For an experimental analysis, the same categories of information can be considered without knowing the generating model: fitted curves, complementary metrics, cross-validation as a sensitivity diagnostic, residual-versus-time and Q–Q plots, parameter uncertainty, and scientific plausibility can all contribute to the rationale for a user-selected model.

C Worked Example: Sampling Times, Noise, and Model Uncertainty

C.1 Purpose and setup

How well can a fitted curve recover behavior between observations? This example uses a known synthetic decay to explore how time-point placement, measurement noise, and replicate count affect that recovery. It complements model comparison: even when the correct curve form is available, the observations may provide too little information to determine its shape reliably.

The synthetic fraction remaining follows the normalized logistic model,

$$F(t) = \frac{1}{1 + \exp((t - t_{50})/s)}, \qquad t_{50} = 50, \quad s = 8.$$

Here, t_{50} is the time at which half remains, and s controls the transition width. The upper and lower asymptotes are fixed at 1 and 0; t_{50} and s are fitted using the app's model-fitting machinery. Time is in illustrative units.

Each sampling condition was simulated 100 times with independent Gaussian measurement errors and fitted with the same logistic model. Standardized errors were shared across conditions to aid comparison. A separate simulated study supplies the illustrative observations. Each replicate is an independent measurement at its assigned time; noise also applies at time zero. The response scale is fixed, rather than obtained by dividing by a noisy initial observation, and noisy observations are not clipped to the range 0–1.

Table 20: Sampling conditions. Panel letters refer to Figure 5. Noise SD is on the fraction-remaining scale.

Panels	Condition	Times	Replicates/time	Noise SD
A, B	Reference	0, 20, 40, 100	1	0.03
C	Misplaced focus	0, 10, 20, 100	1	0.03
E	Transition focus	0, 40, 60, 100	1	0.03
G	More time points	0, 20, 40, 60, 80, 100	1	0.03
D	Higher noise	0, 20, 40, 100	1	0.09
F	More replicates, higher noise	0, 20, 40, 100	3	0.09
H	More times, higher noise	0, 20, 40, 60, 80, 100	1	0.09

The reference is a deliberately simple, sparse schedule, resembling an in vivo time course with several early observations and fewer later observations. Its times are illustrative, rather than a recommended protocol.

C.2 What the fitted curves show

A convincing individual fit can hide weak support between observations. The reference fit in Figure 5A follows the observed points closely. Across repeated studies, however, the fitted transition varies substantially (B). Moving the early observations farther from the transition ("misplaced focus," C) leaves even greater freedom in when and how rapidly the decline occurs. Agreement with the available observations does not establish the behavior within the long unsampled interval.

Moving a time point can be as useful as adding time points. Replacing time 20 with time 60 places observations on both sides of the main transition (E). Recovery improves markedly with the same number of observations. Adding times 60 and 80 to the reference also improves recovery (G). Both schedules track the true curve closely, with much less variation between studies. Their success reflects coverage of the changing part of the curve, rather than simply a larger sample count.

Replicates help with noise, but leave gaps in time coverage. Higher measurement noise worsens recovery under the reference schedule (D). Additional replicates at those same times improve recovery somewhat (F), but substantial uncertainty about the transition remains. Adding informative times instead gives more consistent recovery (H), despite using fewer total observations than the triplicate schedule. Better coverage also partly offsets the increased noise when compared with the original reference (B). These results show the importance of placement in this example; they do not imply that replication or measurement precision is generally unimportant.

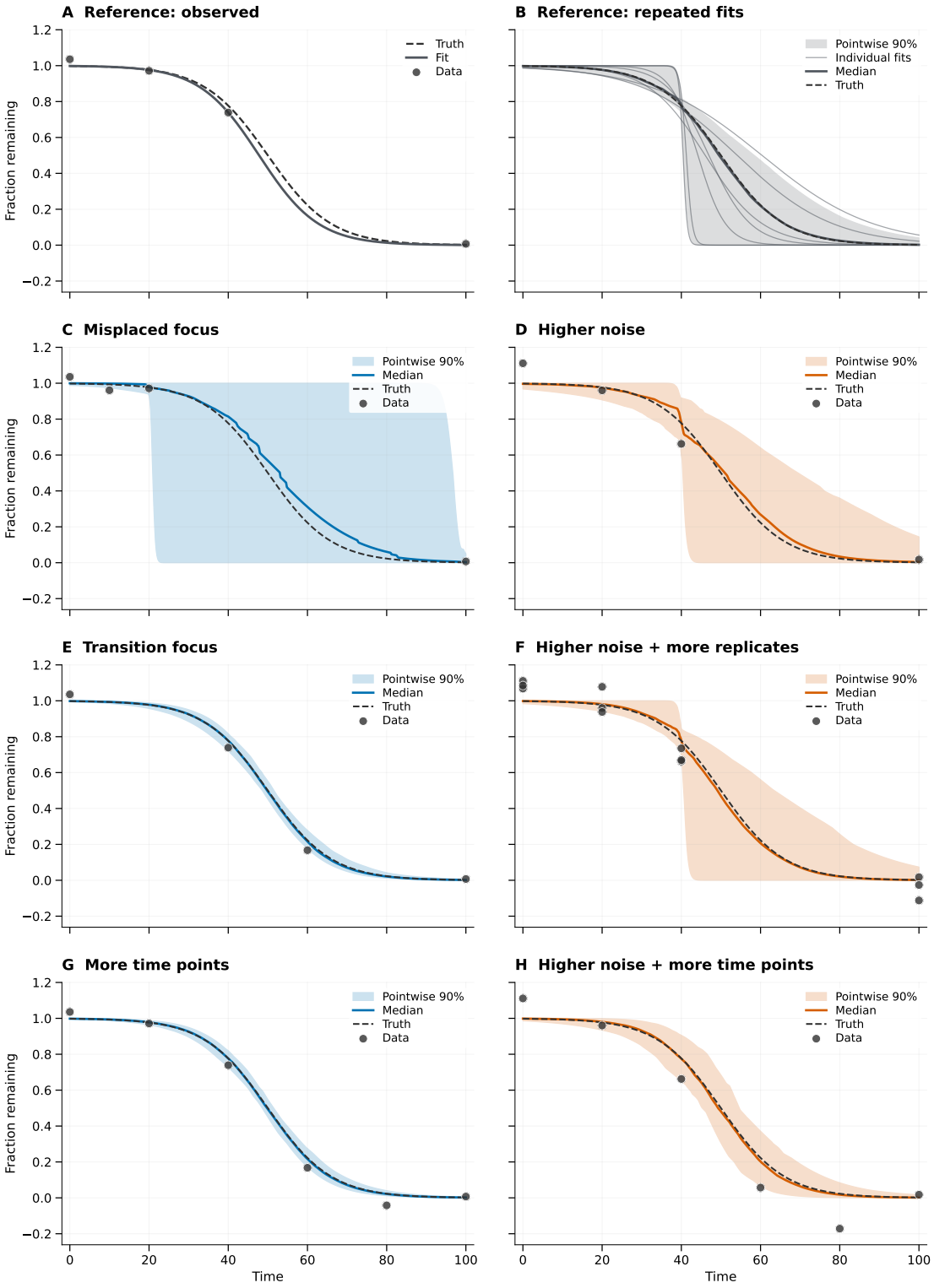


Figure 5: Recovery of the known logistic decay under different sampling conditions. A shows one study and its fitted curve. In B–H, the line and shaded area show the pointwise median and middle 90% of fitted curves across 100 repeated studies. Points in C–H come from the separate illustrative study. B also shows selected individual fits spanning the observed shape variation; these are not frequency-weighted examples. The shaded areas describe variation between simulated studies, not a confidence interval, prediction interval, or cross-validation band from one dataset.

C.3 Recognizing weak support in the app

Figure 6 summarizes the approximate 95% confidence-interval half-width for t_{50}, the same type of parameter uncertainty reported by the app. The reference and misplaced-focus schedules often give wide or unavailable intervals. Transition-focused and denser sampling give much narrower intervals; denser sampling generally narrows them further. Under higher noise, adding times gives more consistently narrow intervals than adding replicates at the reference times. The latter still produces some extremely wide intervals, even though a numerical interval can be calculated.

These are local approximations based on fitted-parameter covariance, and can be unreliable with sparse data. The boxplots summarize only available intervals; unavailable intervals in the reference, misplaced-focus, and higher-noise cases can also signal weak support. A narrow interval alone does not establish that the curve form is appropriate.

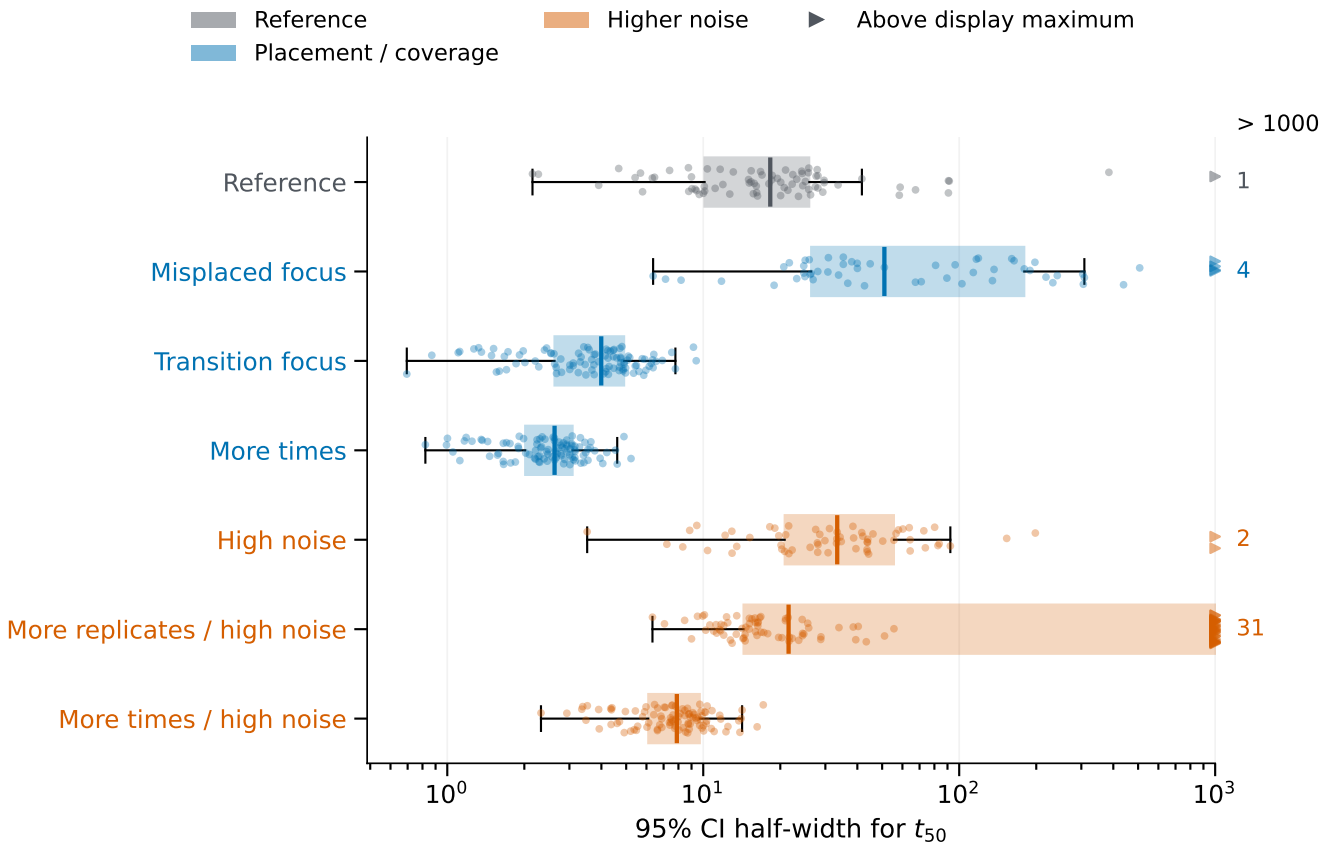


Figure 6: Uncertainty in the fitted transition time. Each point represents an available local 95% confidence-interval half-width for t_{50} from one repeated study; boxes summarize their distribution. Smaller values indicate more precise estimates under the fitted model. The horizontal axis is logarithmic. Right-edge triangles and counts mark values above the display limit; unavailable intervals are excluded.

For an actual dataset, inspect where observations fall relative to the fitted transition, the uncertainty in parameters describing its timing and width, and how the curve changes when observations or time points are omitted during cross-validation. Large parameter uncertainty or strong sensitivity to omissions can motivate additional sampling. Cross-validation cannot supply observations within an unsampled gap, and its variability is distinct from the repeated-study bands shown here.

The favorable transition-focused schedule uses knowledge of where the synthetic transition occurs. In practice, prior information, pilot observations, or broader initial coverage can help locate important changes. Comparing candidate schedules across several plausible transition times and curve shapes would provide a stronger basis for study planning. This example illustrates that reasoning for one known curve and error model; it does not identify a universally best schedule.

D Glossary

The following definitions apply specifically within the IVIVC app and this user guide. They are intended as operational quick references for tool users and do not constitute official FDA regulatory definitions. See Appendix A for more detailed interpretation guidance.

Term	Meaning in this app and guide
Analysis approach	The overarching method used to establish or evaluate the in vitro/in vivo relationship. An approach can use fitted mathematical models or, for Approach 3, direct interpolation-based mappings.
Approach 1: Direct Correlation	Relates in vitro and in vivo times at common response values and fits a model that maps in vitro time to predicted in vivo time. Interpolation by response value can be used when exact response-value matches are unavailable.
Approach 2: Independent Model Fitting	Fits the same model family separately to the preprocessed observed in vitro and in vivo datasets, then compares or applies the fitted relationships. Interpolated points are not used for fitting.
Approach 3: Direct Mapping	Uses response-at-shared-time and time-at-shared-response-value mappings between the in vitro and in vivo datasets rather than fitting a parametric model.
Candidate model	A mathematical form considered for fitting within an applicable analysis approach. Candidate models can differ in shape, number of parameters, and flexibility.
Confidence interval (parameter)	Approximate range used to summarize uncertainty in a fitted parameter. The app reports each parameter as the estimate $\pm$ the half-width of a 95% confidence interval when the required covariance information is available.
Covariance matrix	Matrix calculated from the fitted model that summarizes approximate parameter variances and how parameter estimates vary together. It is used for parameter standard errors, confidence intervals, and parameter-correlation diagnostics.
Cross-validation (CV)	Prediction-error and sensitivity diagnostic in which part of the fitting data is set aside, the model is fit to the remaining observations, and predictions are compared with the omitted observations. CV does not estimate parameter uncertainty.
Densified interpolation	Interpolation option that adds intermediate grid points in both the time and response-value directions before interpolation. The added values are estimates, not additional experimental observations.
Empirical support	Relative support provided by the observed data for one candidate model compared with others under a selected information criterion. It does not establish that a model is correct or suitable for a particular use.
Extrapolation	Use of a fitted model to estimate behavior outside the range of the observed data. Extrapolated behavior generally has greater uncertainty and depends more strongly on the assumed model form.
Fit/CV ratio	Ratio of a full-fit metric value to the corresponding mean cross-validation metric reported by the app. Values near 1 indicate similar full-fit and cross-validation performance; interpretation also depends on the metric and CV scheme.
Fitted model	A candidate model after its parameters have been estimated from the data.
Fitted parameter diagnostics	App summaries that flag large relative standard errors, strong parameter coupling, or combined identifiability concerns for closer review. They do not determine whether a model is valid or acceptable.
Fitting dataset	Required dataset containing paired in vitro and in vivo time-course data used to establish the IVIVC relationship.
Full fit	Model fit obtained using the complete fitting dataset rather than a training subset created for cross-validation.
Goodness of fit	Degree to which a fitted model describes the observations used to estimate its parameters. The app summarizes goodness of fit with selected metrics and fitted-curve displays.

Term	Meaning in this app and guide
Grid search initialization	Procedure that evaluates many candidate parameter starting values before the final nonlinear least-squares fit and selects a promising starting point. It can improve fitting robustness but does not guarantee a global optimum.
Identifiability	Degree to which the available data distinguish among possible values of fitted model parameters. Poor identifiability occurs when substantially different parameter combinations can produce very similar fitted curves.
Information criterion	Model-comparison measure, such as AIC, AIC_c, or BIC, that combines fit quality with a penalty for model complexity. Information-criterion values are interpreted comparatively, not as absolute fit scores.
Interpolation	Estimation of response values between measured points. Interpolated values are derived from the observed data and are not new measurements.
In vitro	Measurements or behavior observed in a laboratory or other setting outside the living organism. In this app, in vitro data provide one side of the IVIVC relationship.
In vivo	Measurements or behavior observed in a living organism. In this app, in vivo data provide the other side of the IVIVC relationship.
IVIVC (in vitro–in vivo correlation)	Quantitative relationship between in vitro and in vivo degradation behavior. The app provides several approaches for establishing, evaluating, and applying such relationships.
Model complexity	Flexibility introduced by a model's form and number of fitted parameters. Additional complexity can improve in-sample fit but can also increase parameter uncertainty or overfitting.
Model parameter	Numerical quantity in a model equation that is estimated from the data during fitting, such as an amplitude, rate constant, exponent, or lag parameter.
Normalization	Preprocessing that expresses response values relative to a reference, such as the first response value or the observed minimum and maximum, so datasets can be compared on a common relative scale.
Observed-response inverse mapping	Approach 2 prediction method that maps each observed in vitro prediction response to the time at which the fitted in vivo calibration curve reaches that response. The app restricts the inverse to a one-to-one fitted curve over the observed in vivo calibration time range and does not extrapolate response values outside the observed in vivo calibration response range. A response within the observed range but just beyond the fitted curve at an endpoint is mapped to that calibration-time endpoint. Approximate 95% inverse-prediction intervals combine fitted in vivo parameter uncertainty with an assumed response uncertainty equal to the in vivo fit's residual standard error.
Overfitting	Situation in which a model's flexibility allows it to follow random variation in the fitted data rather than only the underlying relationship. Overfitting can appear as excellent full-fit metrics accompanied by poorer CV performance or unstable parameters.
Parameter coupling / correlation	Tendency for two fitted parameter estimates to change together. Strong coupling can mean that one parameter can compensate for another while producing a similar fitted curve.
Parameter estimate	Fitted numerical value of a model parameter obtained from the optimization procedure.
Parameter uncertainty	Uncertainty in the numerical values of fitted model parameters. The app summarizes it with approximate confidence intervals, relative standard errors, and parameter-coupling diagnostics where available.
Parametric model	Mathematical model described by an equation with a finite set of fitted parameters. Approaches 1 and 2 can use parametric models; Approach 3 uses direct mappings rather than a fitted parametric model.

Term	Meaning in this app and guide
Prediction band	Model-based interval around a fitted or predicted relationship that incorporates sources of uncertainty defined for the applicable approach. In fitted parametric models, the displayed 95% prediction band generally includes fitted-parameter uncertainty and residual/model error.
Prediction dataset	Optional in vitro dataset used after model fitting to generate corresponding in vivo predictions.
Prediction error	Difference between an observed response and the corresponding model prediction. For fitted observations, this difference is also called a residual.
Preprocessing	Operations applied to uploaded data before fitting, comparison, or prediction. In this app, preprocessing can include normalization, scaling, and interpolation.
Q–Q plot (quantile–quantile plot)	Graph that compares ordered standardized residuals with the quantiles expected from a normal distribution. It is used as a qualitative diagnostic for residual-distribution shape.
Random seed	Starting value used by a pseudorandom procedure so randomized operations, like shuffle split CV and grid search, can be reproduced.
Relative model evidence	Comparison of candidate models using AIC, AIC_c, or BIC, including delta values, weights, and evidence ratios. These quantities describe relative support only within a comparable candidate set.
Relative standard error (RSE)	Parameter standard error expressed as a percentage of the absolute parameter estimate. Large RSE values indicate that uncertainty is large relative to the fitted value.
Residual	Observed response minus the fitted or predicted response (i.e., the error in a given prediction). Residual patterns over time can reveal systematic lack of fit that is not obvious from the fitted curve alone.
Residual degrees of freedom	Number of fitted observations minus the number of estimated model parameters, $n - p$. It is used in estimating residual variance and in the app's approximate parameter confidence intervals.
Residual plot	Plot of residuals against time or another fitted quantity. In this app, raw residual plots show observed-minus-predicted response values versus time in the original response units.
Response value	Measured y-axis quantity that changes with time, such as mass remaining, molecular weight, a mechanical property (e.g., tensile or burst strength), or another quantitative degradation-related endpoint.
Scaling	Preprocessing that changes the numerical scale of time or response values, such as applying a base-10 logarithm, without changing which observations are present.
Scientific plausibility	Consistency of the fitted relationship, parameter values, and predicted behavior with relevant knowledge of the material, degradation mechanism, test conditions, and intended use.
Shared-grid interpolation	Interpolation option that estimates response values at the union of observed time points and estimates times at the union of observed response values. Estimates outside the overlapping range are unavailable.
Standard error (SE)	Approximate standard deviation of a fitted parameter estimate implied by the fitted model and covariance calculation. Smaller values indicate more precise parameter estimates on that parameter's scale.
Test data	Observations set aside during a CV split and used to evaluate predictions from a model fitted without those observations.
Time constant (integral time constant)	Characteristic time scale calculated, when finite, as the area under a fitted decay curve after normalizing its fitted value at time zero to 1, $\tau = f(0)^{-1} \int_0^\infty f(t)\,dt$. In Approach 2, in vitro and in vivo time constants can be compared or used as a ratio for time rescaling.
Training data	Observations retained during a CV split and used to fit the model for that split.

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